HOME ENERGY AUDIT [2]

Energy Audit Report and Carbon Audit of a Shared Student-Hostel Room in Hong Kong

SEE8115 Assignment #1: home energy assessment, EMSD-format energy audit report and Sha Tin District scale-up

Room 1003B, 10/F, Hall 13, CityU Lee Shau Kee Student Residence Village, Ma On Shan, New Territories, Hong Kong



Room 1003B - the audited home (twin room, 15 m², window facing NW)

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Submitted with	Completed NELC Home Energy Audit Tool (.xlsx); completed EE-EAes form (.xls); calculation workbook (.xlsx)

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How this report answers the assignment

Assignment requirement	Where it is answered
(1) Audit the household with the NELC Home Energy Audit Tool; complete and submit the Excel sheet	1_SEE6115-8115_Home Energy Audit Tool_COMPLETED.xlsx: room, washroom, kitchen and appliance sheets, plus the NELC Survey and NELC Questionnaire tick-boxes. Summary in Appendix E
(2a) Executive Summary using the EE-EAes2018 sheet as reference; only relevant sections; submit the Excel sheet	Executive summary, EE-EAes tables (Parts 1(A), 1(B)(I), 2 and 3, with the non-applicable CBSI and lift & escalator sections identified); 2_EE-EAes2018 (V.0)_SEE6115-8115_COMPLETED.xls; cell-by-cell entries in Appendix D
(2b) Objective and scope	Sections 1.2 and 1.3; boundary in Section 3
(2c) Equipment/systems operating characteristics	Section 5: inventory, ratings, controls and operating schedules, with photographs
(2d) At least five potential EMOs with energy savings and cost-benefit analysis for each	Section 9: 21 EMOs, each with baseline, saving, cost, payback, NPV, savings-to-investment ratio and cost of saved energy
Historical energy consumption (recommended); assumed use where bills are unavailable	Sections 6.1-6.2 (Octopus AC spend since move-in, January-September 2026) and 6.4-6.7
Pictures of appliances (recommended)	Sections 4, 5 and 8; Appendix B (26 photographs)
(3) Energy per m² and per capita; scale up to the district with Housing Department and other government data	Sections 10.1-10.3 (Sha Tin District)
(3i) Compare the per-m² and per-capita district results	Section 10.3
(3ii) Develop a factor linking energy, household floor area and number of residents	Section 10.4

Abbreviations

Term	Meaning
AC	Air-conditioning / air-conditioner
BEC / BEEO	Building Energy Code / Buildings Energy Efficiency Ordinance (Cap. 610)
C&SD / HA	Census and Statistics Department / Hong Kong Housing Authority (Housing Department)
CBSI	Central building services installation (EAC term)
CLP	CLP Power Hong Kong Ltd, the electricity supplier to Ma On Shan
CO_2e	Carbon dioxide equivalent
CSE	Cost of saved energy (annualised capital cost ÷ annual kWh saved)
CSPF / EER	Cooling seasonal performance factor / energy efficiency ratio (rated capacity ÷ rated input)
DSD / WSD	Drainage Services Department / Water Supplies Department
EAC	Code of Practice for Building Energy Audit (EMSD, 2018 edition)
EE-EAes	EMSD technical form: Executive Summary of an Energy Audit Report
EMO	Energy Management Opportunity (Category I, II or III)
EMSD / EPD	Electrical and Mechanical Services Department / Environmental Protection Department
EUI	Energy Utilisation Index (MJ/m²/yr; kWh/m²/yr = MJ/m²/yr ÷ 3.6)
GHG / GWP	Greenhouse gas / global warming potential
HKO	Hong Kong Observatory

Term	Meaning
LPD	Lighting power density (W/m²)
M&V	Measurement and verification of savings
MiC	Modular integrated construction
NELC	National Energy Leadership Corps (Penn State home energy audit programme and tool)
NPV / SIR	Net present value / savings-to-investment ratio (present value of savings ÷ cost)
PRH / SSF	Public rental housing / subsidised sale flats
SRO	CityU Student Residence Office (hall management)
WELS	Water Efficiency Labelling Scheme (WSD)

Executive summary

This report presents a systematic energy and carbon audit of the auditor's home: **Room 1003B, 10/F**, a 15 m² twin room in **Hall 13, CityU Lee Shau Kee Student Residence Village** (Ma On Shan, New Territories, Hong Kong), together with the household's share of the shared washroom, floor kitchen and hall laundry. It follows the six steps of the EMSD Code of Practice for Building Energy Audit (EAC 2018) [1]. The executive summary uses the structure of EMSD Technical Form EE-EAes (V.2) [3], and greenhouse-gas emissions follow the EPD/EMSD Guidelines for buildings in Hong Kong [2]. The aim is to establish the household's energy consumption profile, find ways to reduce its energy bills and emissions, and scale the results up to Sha Tin District.

Table ES.1: Key results (typical 12-month baseline year)

Indicator	Result
Household electricity (room + allocated shared services)	2,586 kWh/yr (1,293 kWh per person); 52.8% is used in shared facilities
Electricity used inside the 15 m² room	1,220 kWh/yr; room EUI 292.7 MJ/m²/yr (81.3 kWh/m²/yr)
Room air-conditioning baseline (Octopus spend)	811 kWh/yr; independent energy-label check 884 kWh/yr (+9.1%)
Since move-in (January-September 2026)	1,956 kWh household electricity; HK$900 of Octopus AC top-ups
Largest end-uses	Air-conditioning 34.4%, shower hot water 24.5%, cooking 12.1%
GHG emissions (CLP factor 0.34 / default 0.7 kg CO_2e/kWh)	939 / 1,870 kg CO_2e/yr (470 / 935 kg per person)
EMOs identified	21 (10 Category I, 7 Category II, 4 Category III)
No-cost Category I package	660 kWh/yr (25.5%) and 224 kg CO_2e/yr saved, including HK$381 a year of Octopus AC charges
Sha Tin District scale-up (Section 10)	889 GWh/yr per capita; 1,896 GWh/yr per m²; composite factor 1,250 GWh/yr against an EMSD-based 1,222 GWh/yr

Executive summary in the EE-EAes (V.2) structure

EE-EAes is written for commercial buildings audited under Cap. 610. Only the sections that describe a household are completed here and in the submitted form. Items about central building services installations (CBSI) and lifts and escalators cannot exist inside a household boundary, so they are marked N/A, as the form instructs (see the last table in this group). The values come from the calculation workbook and were entered into the official macro-enabled form, whose macros and tick-boxes are preserved.

Table ES.2: EE-EAes Part 1(A) - Administrative information (EAC §8.1(m))

Form item	Entry	Note
Name of building	Room 1003B, 10/F, Hall 13, CityU Lee Shau Kee Student Residence Village (household energy audit)	Audited unit = the household's room
Address of building	Lee Shau Kee Student Residence Village, Ma On Shan, Sha Tin District, New Territories	CityU hall
1) Date of commencement	01/01/2026	Occupation and data collection began
2) Date of completion	11/10/2026	Within 6 months of the reference month
3) Energy Audit Form validity	N/A	Voluntary audit; no form issued under Cap. 610
4) Energy Audit Report reference	SEE8115-HEA-2026-01	

Form item	Entry	Note
5) Energy imported from / exported to other buildings?	No (ticked)	Supplied from the hall's CLP supply

Table ES.3: EE-EAes Part 1(B)(I) - Building type, usage and operation (EAC §8.1(b))

Form item	Entry	Relevance
1)(a) Type of building	Not ticked	The form offers only commercial types; the audited unit is a residential student-hostel room
1)(b) Portion of building entity that is common area	0%	Shared washroom and kitchen are allocated by user (Section 3)
1)(c) No. of blocks	1	
2) Total internal floor area	15.0 m²	Room 1003B
3) No. of floors	1 (on 10/F)	
4) Major type of facade	Non-curtain wall (ticked)	MiC concrete modules with aluminium windows
5) Date of occupation approval	Q3 2024 (residence opened)	Exact date not published
6) Type of central air-conditioning	Not applicable (ticked)	Unitary window AC and kitchen split AC only
7)-8) Operation of CBSI-served areas	N/A	No CBSI-served areas in a household
For information: operation of the audited unit	2 occupants; 103 occupied h/week; 7.5 m² per person	EAC §8.1(b) occupant density and hours

Table ES.4: EE-EAes Part 2 - Historical energy consumption analysis (EAC §8.1(g)-(h))

Form item	Past 1st 12-month	2nd / 3rd	Note
1) Annual electricity (kWh/annum)	1,219.6	N/A	Room 1003B; occupied from Jan 2026, so the 12 months to Sep 2026 are a typical year
2) Other energy (MJ/annum)	0	N/A	No Towngas or LPG
3) Annual total energy (MJ/annum)	4,391	N/A	Form formula: 1) × 3.6 + 2)
4) Annual EUI (MJ/m²/annum)	292.7	N/A	= 81.3 kWh/m²/annum
6) Breakdown (MJ/annum): AC / lighting / lift & escalator / others	3,234 / 420 / N/A / 736		Lift & escalator not relevant to a household
7) Energy supply from CBSI (%)	N/A		No CBSI; the share used in shared hall facilities (52.8%) is reported instead (Section 6.8)
8) Energy bill reference month	30/09/2026		September 2026 (Octopus pre-paid AC; no utility bill)

Table ES.5: EE-EAes Part 2 item 5 - Monthly EUI of the past 1st 12-month period (MJ/m²/month)

1: Oct	2: Nov	3: Dec	4: Jan	5: Feb	6: Mar	7: Apr	8: May	9: Jun	10: Jul	11: Aug	12: Sep
32.23	16.60	16.87	16.87	16.06	16.87	16.60	32.23	31.96	32.23	32.23	31.96

Table ES.6: EE-EAes Part 3 - Energy Management Opportunities entered in the form (EAC §8.1(l)(i))

Slot	Ref	Cat.	Type	Energy Management Opportunity	kWh/yr saved*	Payback
A	EMO-01	I	AC	Raise the AC set-point from 25 °C to 26 °C and keep the 20 W desk fan running for air movement	24	Immediate

Slot	Ref	Cat.	Type	Energy Management Opportunity	kWh/yr saved*	Payback
B	EMO-02	I	AC	Use the AC's built-in timer to switch off 2 h before waking in hot months (12 to 10 h/day)	100	Immediate
C	EMO-04	I	AC	Cool-season 'fan first': no AC cooling in December-February (HKO mean 16.5-18.2 °C); use the desk fan and open window, Dry mode only on humid days	107	Immediate
D	EMO-05	I	LG	Use the desk task lights instead of both ceiling battens for 3 of the 6 lit hours	44	Immediate
E	EMO-06	I	Others	Shorten showers from 5 to 4 minutes (phone timer or a 4-minute song)	127	Immediate
F	EMO-08	I	EL	Laptop power management: lower brightness, sleep after 10 min idle, battery-saver mode	28	Immediate
G	EMO-10	I	Others	Air-dry laundry on a rack by the open window instead of tumble-drying (keep the dryer for bulky items in humid spells)	50	Immediate
H	EMO-11	II	AC	Washroom exhaust fan: replace 24 h running with an occupancy / humidistat run-on control (~8 h/day)	175	3.2 yr
I	EMO-12	II	AC	Kitchen AC: 7-day timer + occupancy interlock, set-point locked at 25.5 °C (8 to 5 h/day hot, 4 to 2 h/day cool)	524	1.4 yr
J	EMO-14	II	Others	Fit a WELS Grade-1 low-flow showerhead (9 to 7 L/min) in the shared washroom	282	0.6 yr
K	EMO-18	III	Others	Replace 24.4 kW instantaneous shower heaters with central air-source heat-pump water heating (COP about 3.5) and storage	910	15.6 yr
L	EMO-21	III	Others	Expand rooftop solar PV under the CLP Feed-in Tariff (the village already has 110 panels)	-	-

**Facility level for hall measures. The form has 12 EMO rows: Category I 7, Category II 3, Category III 2; types LG 1, AC 5, EL 1, LE 0, Others 5. The full register of 21 EMOs is in Section 9 (LG 2, AC 9, EL 2, LE 0, Others 8).*

Table ES.7: EE-EAes sections judged not relevant to a household

Form section	What it covers	Why it is N/A here
Part 1(B)(II)(1)(a)(i)-(ii), (1)(b)-(2), (3)-(5)	Central chillers, VRF, air-handling units, pumps, CBSI lighting and lift & escalator inventories	A household has no central building services; the room's own unitary AC and LED lights are reported in Section 5
Part 1(B)(I) items 6-8	Central AC type and operation of CBSI-served areas	Ticked 'Not applicable' / entered N/A
Part 2 item 6 (lift & escalator) and item 7	Lift energy; energy supplied by CBSI to units	Lifts are hall common services outside the boundary; no CBSI supply to the unit
Part 3 type 'LE'	Lift/escalator EMOs	None possible within the household

Main findings

- **Operation, not equipment, drives use.** The room AC (Midea MW-12CRF8E) and the kitchen appliances already carry Grade 1 HK Energy Labels. Energy is driven by all-night AC operation at 25 °C, AC use in the cool months, a washroom exhaust fan that runs 24 hours a day and a kitchen AC that residents leave on for 8 h or more.
- **More than half of the "home" is outside the room.** 52.8% of the household's electricity is used in shared facilities that students do not control, chiefly shower water heating (24.4 kW instantaneous heater), the washroom fan and the kitchen.
- **Hot water is seasonal.** Shower energy is 2.12 times higher in January than in July because the cold-water inlet falls from about 29 °C to 16.5 °C [8].

- **Lighting is already efficient.** It is all-LED, and the room's lighting power density of 4.0 W/m² is within the BEC 2021 dormitory limit of 6.2 W/m² [24].
- **The emission factor doubles the answer.** Scope 2 is 93.6% of the inventory; the CLP factor (0.34) and the territory default (0.7) differ by a factor of two, so both are reported [2], [5].
- **District scale-up needs both floor area and residents.** Scaled per person, the audit implies 889 GWh/yr for Sha Tin; scaled per m², 1,896 GWh/yr. The 2.13× gap equals the ratio of floor area per person (16.0 vs 7.5 m²). A composite factor of 61.8 kWh per m² plus 829 kWh per resident gives 1,250 GWh/yr, within 2% of an EMSD-based benchmark.

Recommendations

- Implement the ten no-cost **Category I** measures now (EMO-01 to EMO-10):
  - raise the AC set-point to 26 °C with the fan on, and stop AC cooling in December-February;
  - use the AC timer to stop 2 h before waking, and clean the filter fortnightly;
  - use task lighting, take 4-minute showers at 38 °C, use laptop power management, cook efficiently and air-dry laundry.

  Together they save about 660 kWh/yr (25.5%).
- Ask the Student Residence Office to implement five low-cost **Category II** measures (EMO-11 to EMO-15): exhaust-fan run-on control, a kitchen-AC timer with a 25.5 °C set-point, dispenser night set-back, WELS Grade 1 showerheads and kitchen occupancy sensors. Together they save about 1,381 kWh/yr at facility level, with paybacks of 0.6-3.2 years.
- Treat central heat-pump water heating and more rooftop PV as **Category III** feasibility studies. Replacing the window AC is **not viable**: its payback of 20.9 years exceeds the 12-year service life.
- Buy a plug-in power meter (about HK$120, within the HK$200 budget) and keep monthly Octopus readings so savings can be verified (EAC §7.6.3).

1 Introduction, objective and scope

1.1 Background

Buildings account for about 90% of Hong Kong's electricity consumption [30]. Electricity generation produced 61% of the territory's greenhouse-gas emissions in 2023, when per-capita emissions were about 4.58 t CO_2e [10]. Hong Kong's Climate Action Plan 2050 targets carbon neutrality before 2050 and aims to cut electricity use in residential buildings by 20-30% from 2015 levels by 2050 [31]. A home is small enough to inventory completely, yet it already raises the problems of a corporate audit: boundaries, shared services, missing meters and the choice of emission factors. The best way to learn carbon auditing is to perform one, and an energy audit is its foundation.

1.2 Objective

The purpose of this audit is to determine the household's own energy consumption profile and identify ways to reduce its energy bills and emissions. The specific objectives are to:

1. establish an energy baseline for the home (annual and monthly, by end-use and by space) and analyse the consumption history since move-in;
2. describe the operating characteristics of the equipment and systems that use energy;
3. quantify Scope 1, 2 and 3 greenhouse-gas emissions using the Hong Kong building guidelines [2];
4. identify, categorise (Category I/II/III) and cost Energy Management Opportunities, with a cost-benefit analysis for each [1];
5. present the results as an EMSD-style energy audit report with an EE-EAes executive summary;
6. derive energy per square metre and per capita, scale them up to Sha Tin District, and develop a factor linking energy, floor area and residents;
7. draw lessons for the forthcoming corporate carbon audit.

1.3 Scope of the audit (EAC §8.1(a))

Installations audited:

- room window AC, LED lighting, desk fan, laptops and phone chargers;
- shared washroom instantaneous water heater, exhaust fan and sensor lighting;
- shared-kitchen refrigerators, water dispenser, split AC, lighting, induction hobs and microwave;
- hall laundry washer and dryer.

Not audited:

- hall common services: lifts, corridors, lobbies and central plant (CBSI);
- the building envelope beyond the room's window;
- other residents' personal use.

Assumptions and estimating methods (detailed in Section 2 and Appendix C):

- room AC energy is the reported Octopus spend divided by the CLP 2026 average tariff, checked against the HK Energy Label;
- other loads are rated power × reported hours, and shower energy is calculated from first principles;
- shared fixed loads are allocated per user; the roommate's personal loads are assumed equal to the auditor's.

1.4 Administrative information

Table 1.1: Administrative information (EAC §8.1(m))

Building	Hall 13, CityU Lee Shau Kee Student Residence Village (opened Q3 2024) [15]
Audited unit	Room 1003B, 10/F - twin room, 2 occupants

Address / location	Ma On Shan, Sha Tin District, New Territories, Hong Kong; 22°26'0" N, 114°14'32" E (44 m elevation, phone compass)
Electricity supplier	CLP Power Hong Kong Ltd (room AC pre-paid by Octopus card)
Auditor	Jayash Sahu (SEE8115); occupant of the room
Audit commenced / completed	01/01/2026 / 11/10/2026
Energy bill reference month	September 2026: 1-30 September 2026 (no utility bill; Octopus pre-paid AC). Completion is within the 6 months that EAC §8.1(m) allows
Registered Energy Assessor	Not applicable: voluntary student audit, not submitted under Cap. 610
Previous energy audit (§8.1(k))	None; this is the first audit of the room
Report reference	SEE8115-HEA-2026-01

1.5 Regulatory and methodological basis

The EAC 2018 [1] defines the audit steps (§7), the EMO categories (§6.2) and the content of an audit report (§8), including the executive summary (§8.2) that EE-EAes formalises [3]. Under Cap. 610, energy audits are mandatory only for commercial buildings and the commercial portions of composite buildings, so this audit applies the Code voluntarily. Emissions follow the EPD/EMSD Guidelines [2], which require Scope 1 and 2 reporting, make Scope 3 optional, and require electricity to be reported with both the territory-wide default and the power company's own factor.

The course's NELC (Penn State) Home Energy Audit Tool [4] was used as the room-by-room data-collection template; a completed copy accompanies this report. Equipment efficiency is assessed against the EMSD Mandatory Energy Efficiency Labelling Scheme [9] and lighting against the Building Energy Code [24]. District data come from the Census and Statistics Department and the Housing Department [33]-[37].

2 Audit methodology

2.1 The six EAC steps applied to a home

Table 2.1: How each EAC 2018 step was carried out

EAC step	Requirement	How it was applied in this audit
1 Building information (§7.2)	Inventories, drawings, operating records, 36-month energy data	48-question occupant questionnaire; 26 photographs of nameplates, labels and spaces; CityU building information [15]; HKO climate normals [8]
2 Review of energy-consuming equipment (§7.3)	Site inspection; record equipment characteristics (§7.3.1); calculate power and energy from ratings and operating records, or measure (§7.3.2)	Walk-through of room, washroom and kitchen; equipment inventory and operating schedules (Section 5); Octopus AC spend; label-based and first-principles calculations
3 Identify EMOs (§7.4)	Evaluate AC, LPD, EUI; maintenance, set-points, behaviour, schedules	Benchmarks (EUI, LPD, water, per-capita use); 13 walk-through findings (Section 8)
4 Cost-benefit (§7.5)	Savings, costs, service life, degradation, energy price	HK$1.406/kWh; indicative costs; simple payback, NPV, SIR and cost of saved energy over the service life with 1%/yr degradation
5 Recommendations (§7.6)	Implementation, M&V, further studies	Action plan, metering plan and memo to hall management (Section 11)
6 Report (§7.7, §8)	Report and executive summary	This report; EE-EAes form; completed NELC tool; calculation workbook (live formulas)

2.2 Calculation methods

(1) Appliance energy: E = P × LF × h × d × n ÷ 1000 (kWh/yr). P is rated power (W), LF the load factor, h hours per day, d days per year and n the quantity.

(2) AC, top-down: E = Σ (monthly Octopus spend) ÷ tariff. The spend is HK$140 in May-Oct and HK$50 in Nov-Apr; the tariff is HK$1.406/kWh (CLP 2026 average net tariff) [6].

(3) AC, bottom-up: E = (label kWh ÷ 1,200 h) × Σ operating hours. The HK label states annual consumption for 1,200 h of operation [9].

(4) Shower water heating: E = V × c_p × (T_shower − T_inlet) ÷ (3,600 × η). V is litres (1 L = 1 kg) and c_p = 4.186 kJ/kg·K; T_inlet is the HKO monthly mean air temperature.

(5) Allocation: E_household = Σ E_item × share. Room items count at 1; washroom fixed loads at 2/4; kitchen fixed loads at 2/32; showers, cooking and laundry at the household's own use.

(6) Energy Utilisation Index: EUI = E × 3.6 ÷ A (MJ/m²/yr) [1], [3].

(7) GHG emissions: activity × emission factor. Scope 2 uses both 0.34 (CLP 2025) [5] and 0.7 kg CO_2e/kWh [2]. Water and sewage use the utility's kWh per m³ × 0.7 (Guideline Tables 8 and 9) [2], [11], [12].

(8) Cost-benefit: simple payback = cost ÷ annual saving; NPV = S × [1 − ((1 − d)/(1 + r))^L] ÷ (r + d) − cost, with r = 4% real and d = 1%/yr degradation of savings over the service life L (EAC §7.5.4); SIR = present value of savings ÷ cost; cost of saved energy = cost × r/(1 − (1 + r)^−L) ÷ annual kWh.

(9) District scale-up: district energy = factor × population, × floor area, or (composite) a × floor area + b × residents (Section 10).

2.3 Data sources and data quality

The workbook holds 124 inputs, each tagged with its source and a data-quality class. In descending order of reliability:

- **nameplate or energy-label values photographed on site:** 10;
- **values reported by the occupant in the questionnaire:** 33;
- **published Hong Kong or international data:** 45;

- **derived values:** 3;
- **typical values assumed where nothing better existed:** 33.

Assumed values are highlighted in yellow in the workbook and tested in the sensitivity analysis (Section 12). The most influential assumption is the Octopus AC tariff: CityU does not publish its hall AC rate, so the CLP average tariff is used and cross-checked against comparable university hostels (HK$1.20-2.31/kWh [25]).

2.4 Tools and deliverables

- **Home_Energy_Carbon_Audit_Workbook.xlsx** - 16 sheets with 1,452 live formulas: inputs, AC baseline, hot water, inventory, monthly profile, breakdown, GHG inventory, EMO register with life-cycle costing, sensitivity, district scale-up and the EE-EAes entries.
- **1_SEE6115-8115_Home Energy Audit Tool_COMPLETED.xlsx** - the course's NELC tool completed for the room, washroom, kitchen and laundry, including the NELC Survey and Questionnaire tick-boxes, with all navigation buttons and drop-downs preserved.
- **2_EE-EAes2018 (V.0)_SEE6115-8115_COMPLETED.xls** - the official EMSD form with the relevant sections completed; its macros, tick-boxes and protected layout are unchanged.

3 Audit boundary and reporting period

3.1 Reporting entity and physical boundary

The reporting entity is the **household of Room 1003B**: the auditor and one roommate. The Guidelines ask the reporting entity to define the physical boundary within which emissions are counted [2]. In a student hostel the "home" is not only the room: daily life also depends on a washroom shared by four residents, a floor kitchen used by 32 residents, and hall laundry machines. The boundary therefore combines three kinds of item, shown in the figure below:

- the room in full;
- per-capita shares of the fixed loads in the shared spaces (fans, lights, refrigerators, water dispenser, kitchen AC);
- usage-based items attributed directly (showers, cooking, laundry).

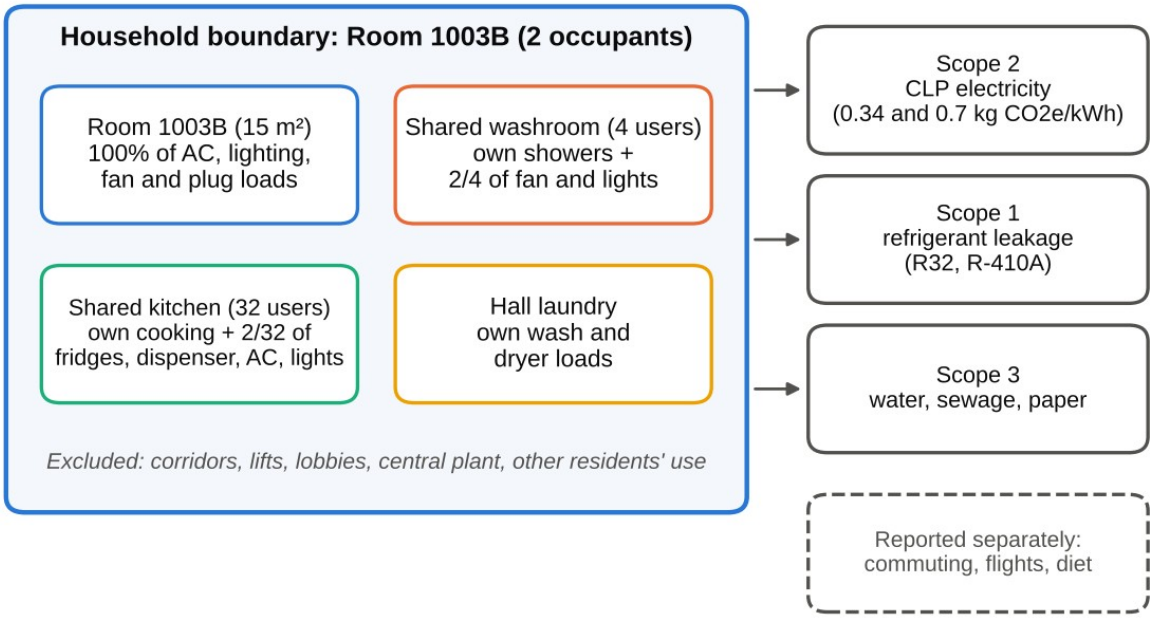


Figure 3.1: Physical and operational boundary of the audit

Per-capita allocation was chosen because every resident has equal access to the shared spaces. Allocating the kitchen over the 20 active users rather than all 32 residents raises the household total by only 4.1% (Section 12). Hall common services (lifts, corridors, lobbies, pumps, central plant and rooftop PV) are outside the boundary because the household cannot influence them and has no data on them.

3.2 Operational boundary

- **Scope 1 - direct:** no fuel is burnt within the boundary; cooking and water heating are all-electric (Guideline Table 1 is nil). Fugitive refrigerant from the room AC (R32) and the kitchen AC (R-410A) is estimated by a screening method (Section 7.2).
- **Scope 2 - energy indirect:** electricity purchased from CLP Power for all items in the boundary (Guideline Table 5). No Towngas is used (Table 6 is nil).
- **Scope 3 - other indirect (optional):** fresh-water supply (Table 8), sewage treatment (Table 9), paper to landfill (Table 7), and seawater for flushing as an extra item.
- **Daily-activity extension:** commuting, flights and diet are reported **separately** (Section 7.6) so that the building inventory stays comparable with the Guidelines.

3.3 Reporting period

The auditor has occupied the room since 1 January 2026, so a 36-month history does not exist. EAC §8.1(g)(i) allows a building operating for less than 36 months to count back to its first day of operation [1]. The nine months since move-in (January-September 2026) are analysed in Section 6.2. Results are expressed as a **typical 12-month baseline year**: the reported hot-season (May-Oct) and cool-season (Nov-Apr) behaviour applied to HKO 1991-2020 climate normals [8]. This is the projection of measured use to a complete operating cycle that EAC §7.5.6(b) allows. The reference month is September 2026. The roughly one-month break reported for 2026 is tested as a sensitivity case (-7%).

4 Site and building characteristics (EAC §8.1(b))

4.1 Location and climate

The residence is in Ma On Shan, New Territories (22°26'0" N, 114°14'32" E, 44 m above sea level), overlooking Tolo Harbour. Hong Kong's humid sub-tropical climate makes cooling the dominant service.

- The HKO 1991-2020 normals (HKO Headquarters, used as representative of the territory) give an annual mean temperature of 23.5 °C [8].
- In the hot season (May-October), monthly means range from 25.7 °C in October to 28.9 °C in July. The coolest month is January, at 16.5 °C.
- Space heating is not provided and was not used.
- The cold season matters for hot water, because the mains-water temperature follows the air temperature.

4.2 The building

Hall 13 is part of CityU's **Lee Shau Kee Student Residence Village** in Ma On Shan [15]:

- 6 halls in 3 towers, housing about 2,100 residents;
- built by modular integrated construction (MiC) from more than 1,300 prefabricated modules, and opened in the third quarter of 2024, with an opening and naming ceremony in December 2024;
- 110 rooftop photovoltaic panels and a building management system.

Each room has its own window AC charged through an Octopus-card system. Lighting and socket power are included in the hall fee.

4.3 The audited room

Table 4.1: Characteristics of Room 1003B

Floor area / occupants	15 m² / 2 (7.5 m² per person; HK median 16 m² [23])
Floor / orientation	10/F; window wall faces NW (328°, phone compass)
Glazing	Single-pane clear glass in a dark aluminium frame with fly screen: 2 casements of 0.53 × 1.26 m (0.67 m² each) + fixed pane of 1.10 × 0.19 m (0.209 m²) = 1.55 m² (10.3% of floor area)
Shading / solar	Full-height grey curtain; direct sun for about 40 min in the late afternoon (NW orientation catches the low summer sun)
Air-conditioning	Midea MW-12CRF8E 1.5 HP inverter window unit above the window (Section 5.1)
Lighting	2 ceiling LED linear battens + 2 under-shelf LED desk lights
Other equipment	2 laptops, 2 phones, 1 desk fan (20 W); no fridge, kettle, heater, TV or router



Photo 02a: NW-facing window and curtain

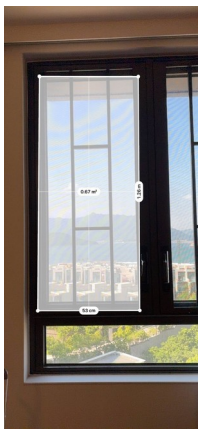


Photo 02c: casement measured 0.53 × 1.26 m



Photo 02d: 328° NW, Ma On Shan

Figure 4.1: Window, glazing measurement and orientation evidence

4.4 Occupancy profile

Table 4.2: Occupancy and operating schedule (reported)

Period	Time in room	Hours	AC use
Weekday	22:00 - 11:00	13 h × 5 = 65 h	On all night in hot months
Saturday / Sunday	Most of the day	20 h + 18 h = 38 h	
Week total		103 h (61% of the week)	
Hot season (May-Oct)		AC 12 h/day at 25 °C	Octopus about HK$140/month
Cool season (Nov-Apr)		AC 4 h/day	Octopus about HK$50/month
Year	About 11 of 12 months in Hong Kong		1-month break (Section 12)

4.5 Shared facilities

- **Washroom** (4 users): Stiebel Eltron DHB-E 27 electronic instantaneous water heater (24.4 kW at 380 V) [16], three LED downlights on an occupancy sensor, and a window exhaust fan that runs continuously.
- **Floor kitchen** (32 residents with access, about 20 active): two Panasonic NR-BV361B refrigerators, an Aquatek AQ-1323T three-temperature water dispenser, two Philips HD4911/80 induction hobs, a Toshiba microwave, and a Toshiba 7.0 kW split AC.
- **Hall laundry:** shared washers and dryers. The auditor uses one cold-wash load (45 min) and 12 minutes of drying per week.

5 Equipment and systems: operating characteristics (EAC §8.1(c))

Table 5.1 lists every energy-using item within the boundary, the evidence behind its rating, how it is used, and its annual energy. Ratings come from nameplates and energy labels photographed on site wherever possible (Appendix B). This fulfils EAC §7.2(b) and §7.3.1; Table 5.2 then describes how each system operates (§8.1(c)).

Table 5.1: Equipment inventory and annual energy (household = Room 1003B)

ID	Item	Make / model / evidence	Use	Share	Household kWh/yr	Per person kWh/yr
R1	Window air-conditioner	Midea MW-12CRF8E, 1.5 HP inverter, R32, Grade 1	Octopus spend ÷ tariff	1	810.8	405.4
R2	Ceiling LED linear battens	2 surface battens (no label visible)	6.0 h/day	1	87.6	43.8
R3	Under-shelf LED desk lights	Integrated in furniture	4.0 h/day	1	29.2	14.6
R4	Oscillating desk fan	ZW02-180, 20 W nameplate	12.0 h/day	1	87.6	43.8
R5	Laptops (auditor + roommate)	HP 65 W adapter TPN-CA17 (avg draw 50%)	8.0 h/day	1	189.8	94.9
R6	Phone chargers	Typical 20 W fast charger	1.0 h/day	1	14.6	7.3
W1	Instantaneous water heater (household showers)	Stiebel Eltron DHB-E 27, 24.4 kW @380 V	See HotWater sheet	1	634.7	317.4
W2	Window exhaust fan (runs 24 h)	J.E.E. window fan (label not visible)	24.0 h/day	2/4	131.4	65.7
W3	LED downlights (occupancy sensor)	3 surface downlights	3.0 h/day	2/4	16.4	8.2
K1	Refrigerators	Panasonic NR-BV361B, Grade 1, 263 kWh/yr label	Energy label	2/32	32.9	16.4
K2	3-temperature water dispenser	Aquatek AQ-1323T, 750 W heater + 160 W compressor	1.5 kWh/day (assumed)	2/32	34.2	17.1
K3	Split air-conditioner	Toshiba RAS-24J2KCV-HK, 7.0 kW, R410A, Grade 1 (688 kWh/1,200 h)	8 h/day hot, 4 h/day cool	2/32	78.7	39.3
K4	Kitchen lighting	Not photographed	16.0 h/day	2/32	32.9	16.4
K5	Induction cooking (household)	Philips HD4911/80, 2,100 W max, Grade 2 (88.1%)	0.4 h/day	1	312.0	156.0
L1	Washing machine (household)	Hall machine, cold wash 45 min	1 load/week each	1	31.2	15.6
L2	Tumble dryer (household)	Hall machine, 12 min per load	12 min/load	1	62.4	31.2
	Total				**2,586.4**	**1,293.2**

EAC §7.3.1 asks for a record of each system's capacity rating, operating characteristics, utilisation pattern and controls, and §8.1(c) asks the report to describe them [1]. Table 5.2 does this for every system in the boundary; its hours are the basis of each kWh in Table 5.1. Three operating patterns stand out:

- **always-on loads** that ignore occupancy: the washroom exhaust fan, the refrigerators and the water dispenser;
- **occupant-switched loads** whose hours follow the occupancy schedule in Section 4.4: the room AC, lights, fan and laptops;
- **high-power, short-duration loads** whose energy depends on behaviour at each use: the 24.4 kW shower heater, the induction hobs and the tumble dryer.

None of the systems has automatic scheduling except the washroom lights' occupancy sensor and the dispenser's cleaning cycle. Metering exists only for the room AC, through Octopus pre-payment (EAC §7.3.1(h)).

Table 5.2: Operating characteristics of the equipment and systems

Equipment / system	Rating, efficiency and control	Operating schedule	Annual operating hours	Household kWh/yr
Room window AC (R1)	Midea MW-12CRF8E inverter, 3.34 kW cooling, 1.192 kW rated input, EER 2.80, CSPF 4.66 (Grade 1); remote control with set-point, Eco, Dry, Timer and fan speed	12 h/day in May-Oct (on all night); 4 h/day in Nov-Apr; set-point 25 °C	2,932 h/yr (12 h × 184 d + 4 h × 181 d); average 0.277 kW while running (23% of rated input)	810.8
Desk fan (R4)	20 W nameplate; 3 speeds and oscillation; manual switch	12 h/day all year	4,380 h/yr	87.6
Ceiling LED battens (R2)	2 × 20 W (assumed); one wall switch; no dimming or sensor	6 h/day (evenings)	2,190 h/yr each	87.6
Desk LED lights (R3)	2 × 10 W (assumed); local switches	4 h/day	1,460 h/yr each	29.2
Laptops (R5)	HP 65 W adapter, efficiency level VI; 32.5 W average in use	8 h/day each; unplugged at night	2,920 h/yr each	189.8
Phone chargers (R6)	20 W each (assumed); unplugged when charged	1 h/day each	365 h/yr each	14.6
Shower water heater (W1)	Stiebel Eltron DHB-E 27: 24.4 kW at 380 V three-phase; electronic temperature control; no storage tank	On demand: 1 × 5-min shower per person per day at 9 L/min, 40 °C	About 61 h/yr for the household; peak 14.9 kW in January	634.7
Washroom exhaust fan (W2)	30 W (assumed); no switch, timer or sensor	24 h/day, every day	8,760 h/yr	131.4
Washroom downlights (W3)	3 × 10 W LED (assumed); occupancy sensor	About 3 h/day	1,095 h/yr	16.4
Refrigerators (K1)	2 × Panasonic NR-BV361B, 332 L, Grade 1, 263 kWh/yr each; thermostat	Continuous; the compressor cycles on its thermostat	8,760 h/yr	32.9
Water dispenser (K2)	Aquatek AQ-1323T: 750 W heater + 160 W compressor; thermostats hold 10 L at 98 °C and a 3 L ice tank; weekly auto-clean by the SRO	24 h/day, every day	8,760 h/yr; standing loss about 1.5 kWh/day (assumed)	34.2
Kitchen split AC (K3)	Toshiba RAS-24J2KCV-HK inverter, 7.0 kW, Grade 1; label average 0.573 kW; remote control, no timer in use	8 h/day in May-Oct, 4 h/day in Nov-Apr (observed left on)	2,196 h/yr	78.7
Kitchen lighting (K4)	6 × 15 W LED (assumed); manual switch	About 16 h/day	5,840 h/yr	32.9
Induction hobs (K5)	Philips HD4911/80: 2,100 W maximum, 5 power levels, Grade 2, 88.1% efficient	3 sessions a week per occupant, 1 h each at about 1.0 kW	312 h/yr (household)	312.0
Hall washer and dryer (L1, L2)	Card-operated hall machines: cold wash about 0.3 kWh/load (assumed); dryer about 3.0 kW (assumed)	1 load a week per occupant: 45-min cold wash and 12 min of drying	Washer 78 h/yr; dryer 21 h/yr	93.6

5.1 Air-conditioning (EAC §8.1(e))

The room AC is a **Midea MW-12CRF8E** 1.5 HP inverter window unit. It is air-cooled (heat is rejected outdoors through the window sleeve) and uses refrigerant R32.

- Rated cooling capacity: 3.34 kW (11,400 Btu/h).
- Rated input: 1.192 kW, giving an EER of 2.80 kW/kW.
- Seasonal performance: CSPF 4.66, HK Energy Label **Grade 1**.

- Label consumption: 362 kWh/yr for 1,200 operating hours [9], [28]. This implies an average seasonal input of only **0.302 kW**, about a quarter of the rated input, because the inverter compressor modulates at part load.
- Controls include Eco, Dry, Timer and fan-speed modes, all usable for EMOs.
- System type (EAC §8.1(e)(iii)): unitary air-conditioners only. There is no central chilled-water plant or air-distribution system within the boundary.

The shared kitchen has a **Toshiba RAS-24J2KCV-HK** 7.0 kW inverter split unit (Grade 1, R-410A, 688 kWh per 1,200 h). Neither unit provides heating.



Photo 03b: Midea window AC - dark deposits on the discharge louvres

Energy Label Information

- Energy Label Grade: 1
- Place of Manufacture: China
- Category: 1
- Refrigerant: R32
- Annual Energy Consumption Cooling (kWh): 362
- Rated Power Consumption Cooling (kW): 1.192
- Rated Cooling Capacity (kW): 3.340
- Cooling Capacity (kW): 3.42
- Cooling Seasonal Performance Factor (CSPF): 4.6618
- Inverter: Yes

Photo 03a (energy-label part): Grade 1, R32, 362 kWh, 1.192 kW, CSPF 4.66

Figure 5.1: Room air-conditioner and its rating

5.2 Lighting (EAC §8.1(f))

The room has two surface-mounted LED linear battens and two under-shelf LED desk lights. No label was visible, so 20 W per batten and 10 W per desk light are assumed.

- Total room lighting power: **60 W**, a lighting power density (LPD) of **4.0 W/m²**.
- This is 35% below the BEC 2021 limit of 6.2 W/m² for dormitories (6.1 W/m² in BEC 2024) [24]. The BEC applies to prescribed buildings, so it is used here only as a benchmark.
- The washroom's three LED downlights are switched by an occupancy sensor (about 3 h/day).

5.3 Plug loads and the desk fan

The auditor's laptop uses an HP 65 W adapter (TPN-CA17, efficiency level VI, which limits no-load losses). Average draw while in use is assumed to be 50% of the adapter rating (32.5 W). Both occupants are assumed to use a laptop for 8 h/day and charge a phone for 1 h/day. The oscillating desk fan is rated at 20 W (nameplate) and runs 12 h/day. The occupants report unplugging devices at night, so standby loads are negligible.



Photo 06a: HP 65 W adapter, efficiency level VI



Photo 06b: desk fan label, 20 W

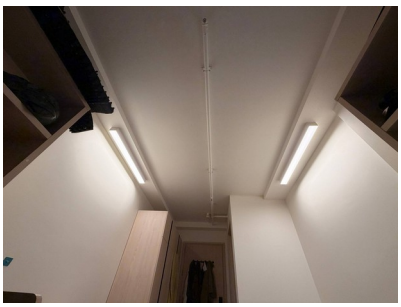


Photo 05a: ceiling LED battens

Figure 5.2: Plug-load and lighting evidence

5.4 Water heating and washroom ventilation

Showers are heated by a **Stiebel Eltron DHB-E 27** electronic instantaneous heater: 27 kW at 400 V, or 24.4 kW at Hong Kong's 380 V three-phase supply [16]. It has no storage tank, so there are no standing losses. However, it draws a very large instantaneous power: a 9 L/min shower in January needs **14.9 kW** (capacity headroom 1.64 ×). Across hundreds of washrooms, this drives the hall's maximum demand. The window exhaust fan (J.E.E.) runs 24 hours a day regardless of occupancy.



Photo 08a: Stiebel Eltron DHB-E instantaneous heater

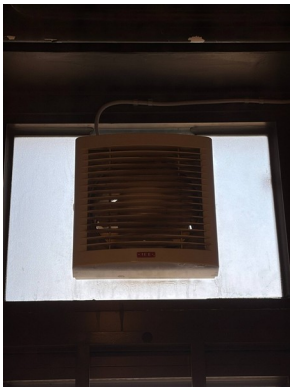


Photo 08d: exhaust fan running 24 h/day

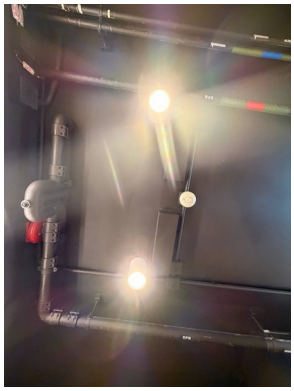


Photo 08b: sensor-controlled LED downlights

Figure 5.3: Shared washroom services

5.5 Kitchen and laundry appliances

The kitchen appliances are efficient, but some run all the time.

- **Refrigerators:** two Panasonic NR-BV361B units (233 L fresh + 99 L frozen), each Grade 1 at 263 kWh/yr.
- **Induction hobs:** Philips HD4911/80, Grade 2, 88.1% thermal efficiency, 2,100 W maximum with five power levels.
- **Water dispenser:** Aquatek AQ-1323T, 750 W heater + 160 W compressor. It keeps 10 L of water at 98 °C plus a 3 L ice tank, 24 hours a day, so it has unavoidable standing losses (assumed 1.5 kWh/day; the ENERGY STAR limit for storage hot-and-cold coolers is 0.87 kWh/day [20]).
- **Kitchen AC:** reported to be left running for 8 h or more by residents (Finding F4).

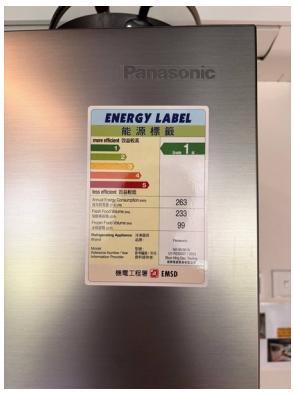


Photo 09i: refrigerator label, Grade 1, 263 kWh/yr

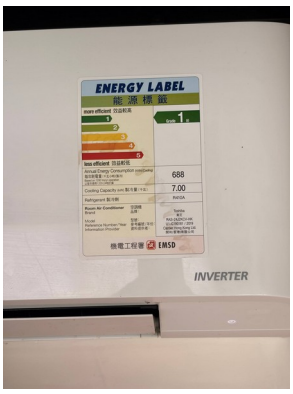


Photo 09g: kitchen AC label, 688 kWh/1,200 h

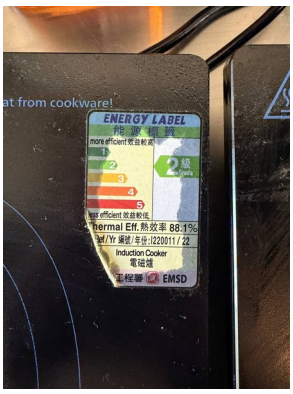


Photo 09e: induction hob, Grade 2, 88.1%

Figure 5.4: Energy labels of shared kitchen appliances

6 Energy consumption analysis (EAC §8.1(d), (g))

6.1 Room air-conditioning baseline: two independent methods

Only the room AC is metered, through Octopus pre-paid charges. The occupants report spending about HK$140 a month in hot months and HK$50 in cool months, HK$1,140 a year for both. Dividing by the CLP 2026 average net tariff of HK$1.406/kWh [6] gives a **top-down baseline of 811 kWh/yr**.

The bottom-up check takes the label's average input (0.302 kW) and multiplies it by the reported pattern of 12 h/day in May-Oct and 4 h/day in Nov-Apr (2,932 h/yr). This gives **884 kWh/yr, +9.1%** relative to the spend-based figure (Table 6.1, Figure 6.1). Agreement within 10% between a financial record and an engineering estimate is good evidence that both the reported behaviour and the baseline are realistic.

Table 6.1: Room AC energy - top-down vs bottom-up

Month	Season	Octopus spend	Top-down kWh	Hours	Bottom-up kWh
Jan	Cool	HK$50	35.6	124	37.4
Feb	Cool	HK$50	35.6	112	33.8
Mar	Cool	HK$50	35.6	124	37.4
Apr	Cool	HK$50	35.6	120	36.2
May	Hot	HK$140	99.6	372	112.2
Jun	Hot	HK$140	99.6	360	108.6
Jul	Hot	HK$140	99.6	372	112.2
Aug	Hot	HK$140	99.6	372	112.2
Sep	Hot	HK$140	99.6	360	108.6
Oct	Hot	HK$140	99.6	372	112.2
Nov	Cool	HK$50	35.6	120	36.2
Dec	Cool	HK$50	35.6	124	37.4
Year		**HK$1,140**	**810.8**	**2,932**	**884.5**

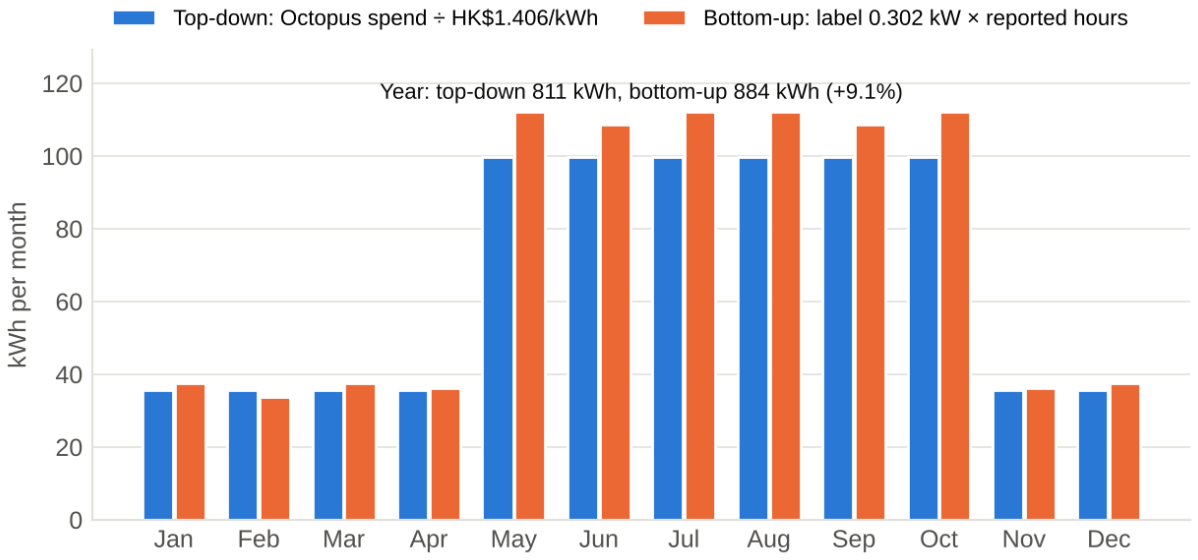


Figure 6.1: Monthly room AC energy, top-down vs bottom-up

What the reconciliation shows:

- If the bottom-up figure were exact, the implied Octopus rate would be HK$1.29/kWh. That lies between the CUHK hostel rate (HK$1.20) and the CLP average (HK$1.406), consistent with an at-cost charge [25].

- Average input while running is only 0.277 kW, 23% of the rated input. This confirms that the inverter mostly runs at part load in a small, lightly loaded room. It is equivalent to 680 full-load hours a year.
- 73.7% of AC energy falls in May-October, so hot-season measures matter most.
- The occupants' first estimate of "about HK$120 a month" would imply 1,024 kWh/yr. It is not consistent with their more specific seasonal figures, so it is treated as an upper bound (+8% in Section 12).

6.2 Historical consumption since move-in (January-September 2026)

The household moved in on 1 January 2026, so the history covers nine months; EAC §8.1(g)(i) allows a shorter history to count back to the first day of occupation [1]. The only metered record is the Octopus pre-payment for the room AC. The occupants report topping up about HK$50 a month in January-April and HK$140 a month from May, **HK$900 in total**. No other load is billed to the room, so each month's other end-uses were estimated with the method of Section 2.2, using that month's number of days and HKO mean temperature (Table 6.2).

- The household used about **1,956 kWh** in the nine months: 194 kWh a month in the cool months (January-April) and 236 kWh a month in the hot months (May-September).
- The step up in May is the room AC (35.6 to 99.6 kWh a month), partly offset by shower energy falling from 77.0 kWh in January to 44.9 kWh in May as the mains water warms.
- The room's monthly EUI doubled from about 16.6 to 32.1 MJ/m².
- Adding the typical October-December months gives the 2,586 kWh baseline year used throughout this report (EE-EAes Part 2, "past 1st 12-month period").

The Octopus amounts are the occupants' typical monthly top-ups rather than individual receipts, so the series is smooth by construction. The reported one-month break would lower the actual total; it is tested in Section 12. Keeping each month's top-up receipt from now on would turn this record into a proper monthly baseline for M&V (Section 11.2).

Table 6.2: Household energy since move-in (Octopus AC record and modelled end-uses)

Month	Days	Octopus AC top-up	Room AC kWh	Hot water kWh	Household kWh	Room EUI MJ/m²	kg CO_2e (CLP)
Jan	31	HK$50	35.6	77.0	207.2	16.9	70.5
Feb	28	HK$50	35.6	67.8	188.8	16.1	64.2
Mar	31	HK$50	35.6	67.5	197.7	16.9	67.2
Apr	30	HK$50	35.6	53.9	181.1	16.6	61.6
May	31	HK$140	99.6	44.9	243.6	32.2	82.8
Jun	30	HK$140	99.6	37.1	232.6	32.0	79.1
Jul	31	HK$140	99.6	36.4	235.0	32.2	79.9
Aug	31	HK$140	99.6	37.0	235.7	32.2	80.1
Sep	30	HK$140	99.6	38.4	233.8	32.0	79.5
Jan-Sep 2026	**273**	**HK$900**	**640.1**	**460.0**	**1,955.6**	**227.0**	**664.9**

6.3 Shower hot water

Shower energy is not metered, so it was calculated from first principles. Each occupant takes one 5-minute shower a day at an assumed 9 L/min, which is 45 L a day. The water is heated to 40 °C from a mains temperature taken as the HKO monthly mean air temperature, at 99% heater efficiency.

- The annual-average temperature rise is 16.5 K.
- Household shower energy is **635 kWh/yr**, 0.87 kWh per person per day.
- January uses 2.12 times as much energy as July (Figure 6.2). The cool months therefore partly offset the hot-season AC peak, which flattens the monthly profile.

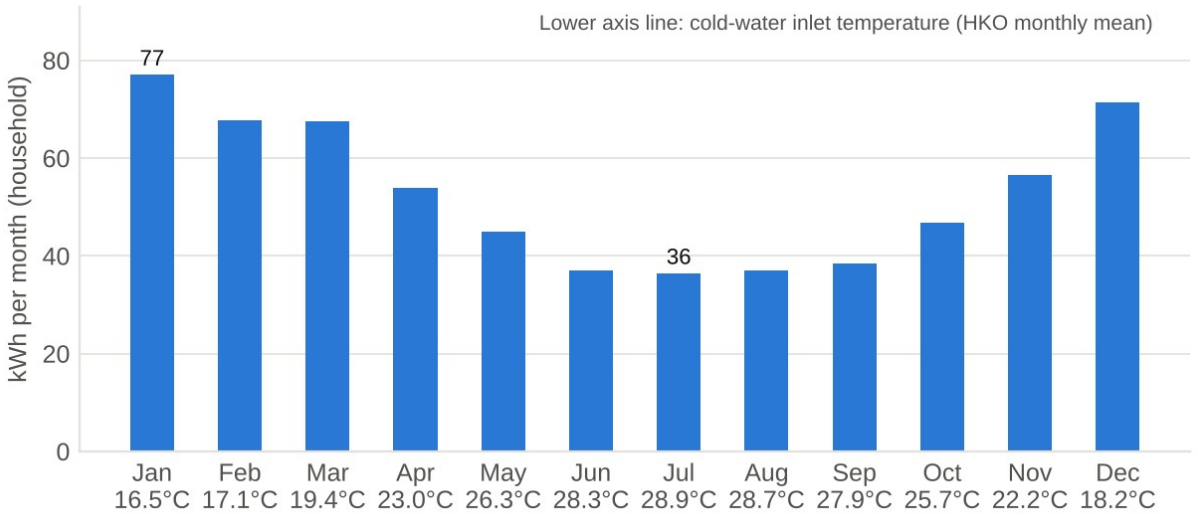


Figure 6.2: Monthly household shower energy and cold-water inlet temperature

6.4 Annual consumption and end-use breakdown

The household uses **2,586 kWh a year** (1,293 kWh per person).

- **Air-conditioning** is the largest end-use at 34.4%: the room AC plus the household's share of the kitchen AC.
- **Shower hot water** is next at 24.5%, then **cooking** at 12.1%.
- Lighting is only 6.4%.
- The washroom exhaust fan's share (5.1%) is larger than all the room's lighting together, even though it is only half of one fan.

Table 6.3: Household electricity by end-use

End-use	kWh/yr	Share	kWh/person/yr	kg CO_2e/yr (CLP)	kg CO_2e/yr (0.7)
Air-conditioning	889.5	34.4%	444.8	302.4	622.7
Hot water	634.7	24.5%	317.4	215.8	444.3
Cooking	312.0	12.1%	156.0	106.1	218.4
Plug loads	204.4	7.9%	102.2	69.5	143.1
Lighting	166.1	6.4%	83.0	56.5	116.3
Ventilation	131.4	5.1%	65.7	44.7	92.0
Air movement (fan)	87.6	3.4%	43.8	29.8	61.3
Laundry	93.6	3.6%	46.8	31.8	65.5
Drinking water	34.2	1.3%	17.1	11.6	24.0
Refrigeration	32.9	1.3%	16.4	11.2	23.0
Total	**2,586.4**	**100.0%**	**1,293.2**	**879.4**	**1,810.5**

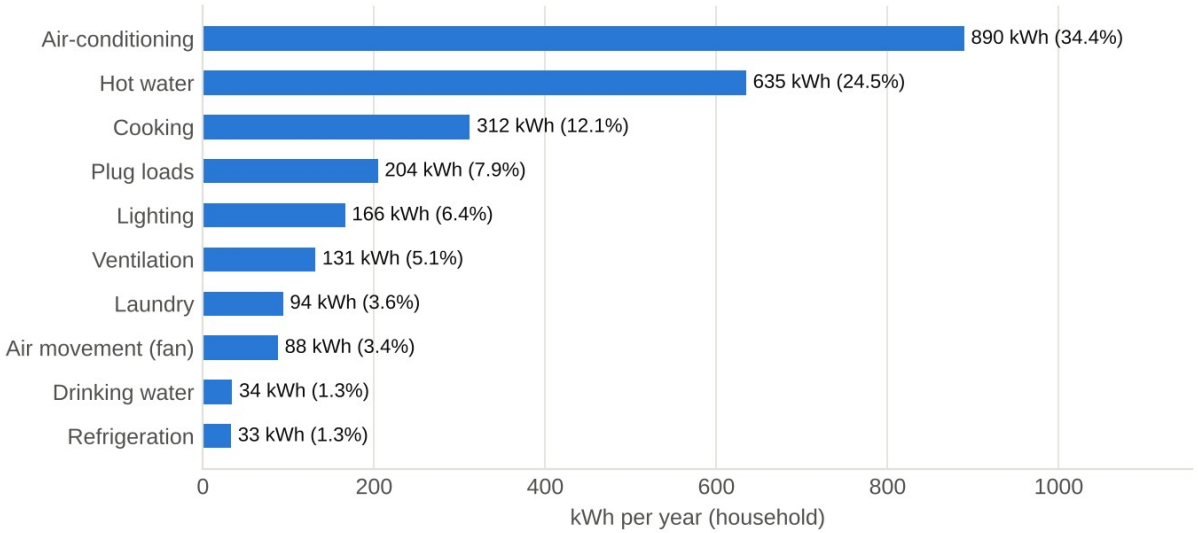


Figure 6.3: Household electricity by end-use (kWh/yr and share)

6.5 Breakdown by EAC installation category (EAC §8.1(g)(iv))

EAC §8.1(g)(iv) asks for the breakdown by installation: air-conditioning, lighting, lift and escalator, and others. The EE-EAes form notes that air-conditioning "includes central mechanical ventilation" [3]. Following that convention, the desk fan and the washroom exhaust fan are counted as air-conditioning.

- **Inside the room:** air-conditioning is 73.7% of energy.
- **For the whole household:** air-conditioning is 42.9%, and "others" grows to 50.7% because of water heating and cooking. Lifts and escalators are outside the boundary.

Table 6.4: EAC categories - room vs household

Category	Room kWh/yr	Room share	Room MJ/yr	Household kWh/yr	Household share
Air-conditioning	898.4	73.7%	3,234	1,108.5	42.9%
Lighting	116.8	9.6%	420	166.1	6.4%
Others	204.4	16.8%	736	1,311.8	50.7%
Lift & escalator	0.0	0.0%	0	0.0	0.0%
Total	**1,219.6**	**100.0%**	**4,391**	**2,586.4**	**100.0%**

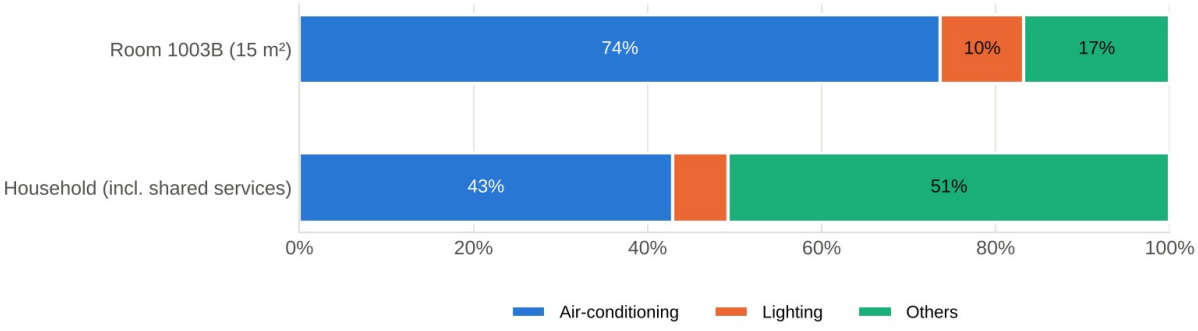


Figure 6.4: EAC installation categories - room vs household

6.6 Monthly profile and monthly EUI (EAC §8.1(g)(iii))

The monthly profile (Table 6.5) is presented graphically in Figures 6.5 and 6.6, as EAC §8.1(g)(v) requires. It is flatter than the AC alone would suggest. In winter, shower energy rises as AC use falls. Household use ranges from 181 kWh in April to 246 kWh in October. Because the room's own loads are dominated by the

AC, the monthly EUI of the room roughly doubles between the cool months (about 16.1 MJ/m²) and the hot months (about 32.2 MJ/m²).

Table 6.5: Monthly energy, room EUI and emissions (typical year)

Month	Days	Season	Room AC	Hot water	Cooking	Other	Household kWh	Room EUI MJ/m²	kg CO_2e (CLP)
Jan	31	Cool	35.6	77.0	26.5	68.2	207.2	16.87	70.5
Feb	28	Cool	35.6	67.8	23.9	61.6	188.8	16.06	64.2
Mar	31	Cool	35.6	67.5	26.5	68.2	197.7	16.87	67.2
Apr	30	Cool	35.6	53.9	25.6	66.0	181.1	16.60	61.6
May	31	Hot	99.6	44.9	26.5	72.6	243.6	32.23	82.8
Jun	30	Hot	99.6	37.1	25.6	70.3	232.6	31.96	79.1
Jul	31	Hot	99.6	36.4	26.5	72.6	235.0	32.23	79.9
Aug	31	Hot	99.6	37.0	26.5	72.6	235.7	32.23	80.1
Sep	30	Hot	99.6	38.4	25.6	70.3	233.8	31.96	79.5
Oct	31	Hot	99.6	46.9	26.5	72.6	245.5	32.23	83.5
Nov	30	Cool	35.6	56.4	25.6	66.0	183.6	16.60	62.4
Dec	31	Cool	35.6	71.4	26.5	68.2	201.7	16.87	68.6
Year	**365**		**810.8**	**634.7**	**312.0**	**828.9**	**2,586.4**	**292.7**	**879.4**

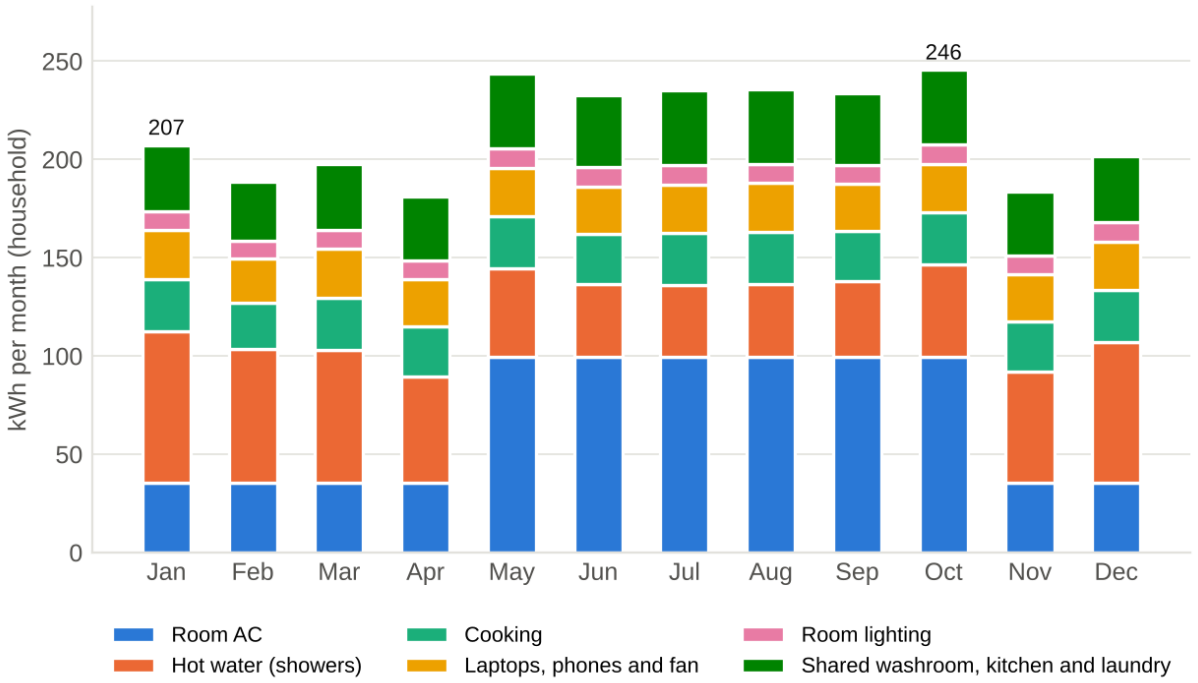


Figure 6.5: Household electricity by month and end-use group

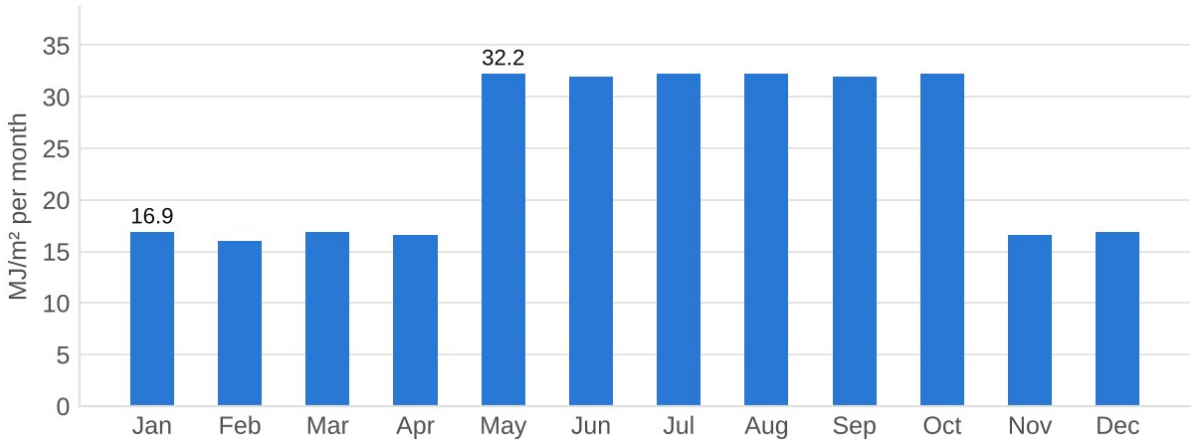


Figure 6.6: Monthly EUI of Room 1003B (MJ/m² per month)

6.7 Energy Utilisation Index and benchmarks

The room's EUI is **292.7 MJ/m²/yr (81.3 kWh/m²/yr)**. Hong Kong does not publish a statutory residential EUI benchmark, so one was derived from public statistics. A typical three-person household uses 275 kWh a month [7], and the median floor area per person is about 16 m² [23]. Together these give about 68.8 kWh/m²/yr (248 MJ/m²/yr).

The room is 1.18 × this benchmark, and household electricity per person is 1.18 × the typical 1,100 kWh. Two caveats apply:

- typical homes often cook and heat water with Towngas, whereas this household is all-electric, so the electricity comparison is conservative;
- the room has half the median floor area per person, so the same AC use is concentrated over fewer square metres.

Water use is well below average (70.6 vs 120.9 L/person/day [13]), and lighting power density is well below the BEC limit (Figure 6.7).

Table 6.6: Key performance indicators vs Hong Kong benchmarks

Indicator	This household	Benchmark	Ratio	Benchmark source
Room EUI (kWh/m²/yr)	81.3	68.8	1.18	275 kWh/month ÷ 3 persons ÷ 16 m² [7], [23]
Electricity per person (kWh/yr)	1,293	1,100	1.18	Typical 3-person household [7]
Fresh water (L/person/day)	70.6	120.9	0.58	HK domestic 2023/24 [13]
Floor area per person (m²)	7.5	16.0	0.47	2021 Census median [23]
Lighting power density (W/m²)	4.0	6.2	0.65	BEC 2021 dormitory [24]

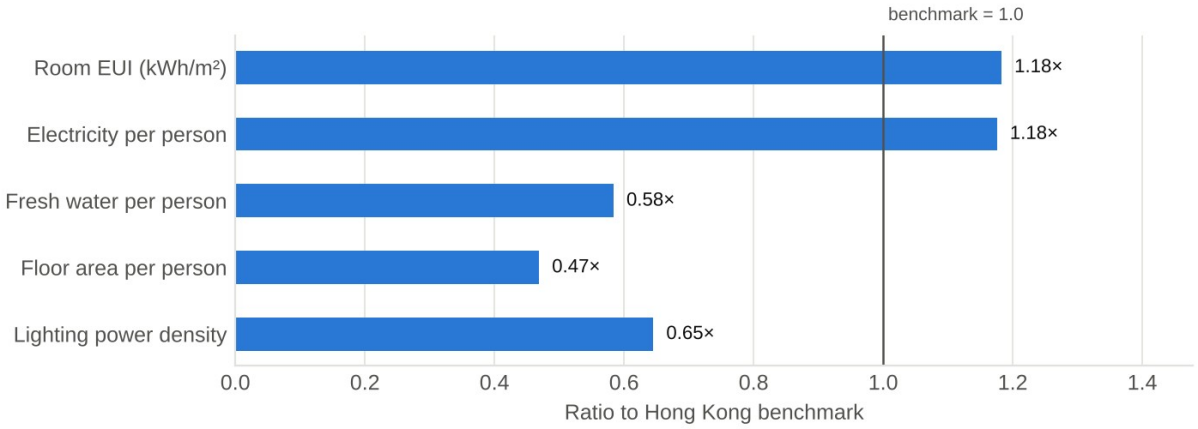


Figure 6.7: Household indicators as a ratio of Hong Kong benchmarks

6.8 Energy used outside the room (EAC §8.1(h))

EAC §8.1(h) asks how much energy central services supply to the building's units. The equivalent here is the share of household energy used in shared facilities: **52.8%**, or 1,367 kWh/yr. It splits into the washroom (783 kWh), the kitchen (491 kWh) and the laundry (94 kWh).

This is the most important structural finding of the audit. A resident who changed only in-room habits could reach less than half of the household's footprint. EAC §7.4.3 allows EMOs that depend on other responsible persons to be recorded and conveyed to them for follow-up [1]. Here the roles are reversed: the audited unit is the room, and the person responsible for the shared services is the hall management. Many EMOs therefore have to be conveyed to the hall management.

7 Greenhouse-gas inventory

7.1 Emission factors

Table 7.1: Emission factors used (Hong Kong-specific where available)

Activity	Factor	Basis and source
Electricity - CLP Power (headline)	0.34 kg CO_2e/kWh	CLP Power HK, electricity sold, 2025 (0.38 in 2024; 0.39 in 2023) [5]
Electricity - territory-wide default	0.70 kg CO_2e/kWh	Guideline Table 5; must also be reported [2]
Fresh water (WSD)	0.437 kg CO_2e/m³	0.624 kWh/m³ (WSD 2023/24) × 0.7 - Guideline Table 8 method [11]
Sewage (DSD)	0.203 kg CO_2e/m³	0.29 kWh/m³ (DSD 2021/22) × 0.7 - Guideline Table 9 method [12]
Seawater for flushing (WSD)	0.253 kg CO_2e/m³	0.362 kWh/m³ × 0.7 (optional item) [11]
Paper to landfill	4.8 kg CO_2e/kg	Guideline Table 7 [2]
Refrigerants (GWP)	HFC-32: 650; R-410A: 1,725	Guideline Table 3-1 [2]

The Guidelines date from 2010, and their tabulated factors (e.g. CLP 0.54 kg/kWh in 2008, water 0.424 kg/m³) are out of date. As the Guidelines themselves instruct, the latest published factors have been used with the Guidelines' methods [2]. CLP's factor has fallen from 0.39 (2023) to 0.34 kg CO_2e/kWh (2025) as its fuel mix has shifted towards natural gas and non-carbon sources [5].

7.2 Scope 1 - direct emissions

Combustion (Guideline Table 1): nil. There is no Towngas, LPG or other fuel, because cooking (induction) and water heating (electric) are all-electric.

Refrigerant (Guideline Table 3): the mass-balance method uses the change in refrigerant stock (purchases, disposals, inventory). The occupants hold no refrigerant, so this method gives zero. That would understate reality, so a screening estimate (IPCC operating leak rate of 4% of charge per year [21]) is added to Scope 1. The charges are assumed, because neither was on the photographed labels (Table 7.2).

Table 7.2: Scope 1 screening estimate of fugitive refrigerant

Equipment	Refrigerant	Charge (kg)	Leak rate	GWP	Household share	kg CO_2e/yr
Room window AC	R32	0.40	4%	650	1	10.4
Kitchen split AC	R-410A	1.40	4%	1,725	2/32	6.0
Scope 1 total						**16.4**

7.3 Scope 2 - purchased electricity

Table 7.3: Scope 2 by facility (Guideline Table 5 format)

Facility / source	kWh/yr	kg CO_2e/yr (CLP 0.34)	kg CO_2e/yr (default 0.7)
Room 1003B (incl. Octopus-metered AC)	1,219.6	414.7	853.7
Shared washroom (own showers + 2/4 of fan & lights)	782.5	266.1	547.8
Shared kitchen (own cooking + 2/32 of fixed loads)	490.6	166.8	343.4
Hall laundry (own use)	93.6	31.8	65.5
Scope 2 total	**2,586.4**	**879.4**	**1,810.5**

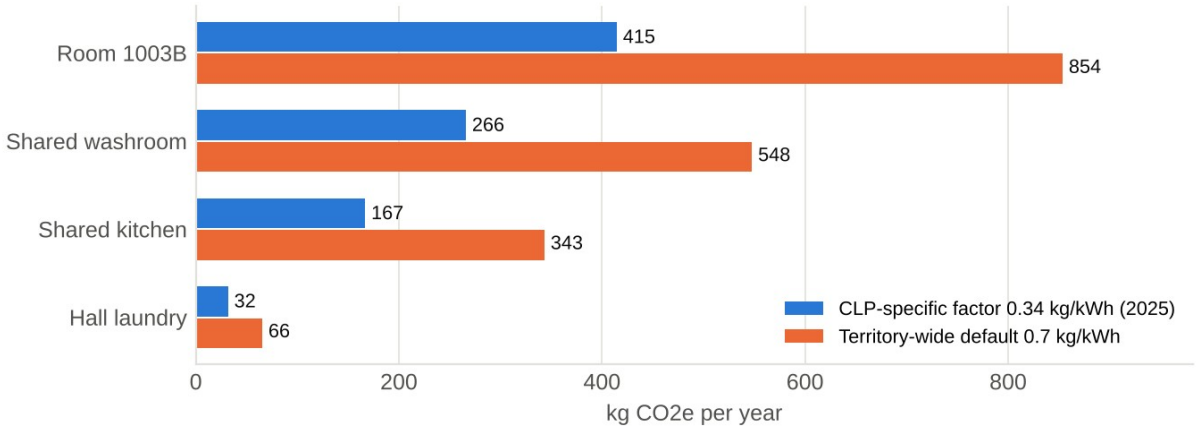


Figure 7.1: Scope 2 emissions by facility under the two required factors

7.4 Scope 3 - water, sewage and paper

Fresh water is estimated bottom-up at 70.6 L per person per day: showers 45 L, drinking water from the dispenser 2 L, basin, cooking and dish-washing about 15 L, and laundry about 8.6 L. The household total is 51.5 m³/yr; all of it is assumed to go to the public sewer.

Toilet flushing uses seawater, which serves about 85% of Hong Kong's population [13]. It is reported separately at 21.9 m³/yr.

Paper: the household uses about 2.0 kg a year. The auditor recycles, and half of the paper is assumed to be recycled. The remainder is counted to landfill at the Guideline factor.

Table 7.4: Scope 3 items (Guideline Tables 7-9 format)

Guideline item	Quantity	Unit	Factor	kg CO_2e/yr
Table 7 - Paper waste to landfill	1.00	kg	4.800 kg CO_2e/kg	4.80
Table 8 - Fresh water (WSD)	51.52	m3	0.437 kg CO_2e/m3	22.50
Table 9 - Sewage (DSD)	51.52	m3	0.203 kg CO_2e/m3	10.46
Seawater for flushing (WSD, optional)	21.90	m3	0.253 kg CO_2e/m3	5.55
Scope 3 total				**43.31**

7.5 Inventory summary

Table 7.5: GHG inventory of the household (kg CO_2e/yr)

Scope	CLP-specific basis	Territory default basis	Share (CLP)	Per person (CLP)	Per person (default)
Scope 1 - direct (refrigerant screening)	16.4	16.4	1.8%	8.2	8.2
Scope 2 - purchased electricity	879.4	1,810.5	93.6%	439.7	905.2
Scope 3 - water, sewage, paper	43.3	43.3	4.6%	21.7	21.7
TOTAL	**939.1**	**1,870.2**	**100.0%**	**469.6**	**935.1**

The household emits **939 kg CO_2e a year** on the CLP-specific basis (470 kg per person), or 1,870 kg on the territory-wide default. Scope 2 electricity is 93.6% of the total; the room's own Scope 2 intensity is 27.6 kg CO_2e/m²/yr. Because Scope 2 dominates, the choice of factor matters more than any single item of equipment.

- The 0.7 default reflects a coal-heavy grid of the 2000s.
- CLP's 2025 factor reflects today's gas and non-carbon mix and is the better estimate of this household's actual emissions.

Reporting both, as required, makes the results comparable with older Hong Kong building inventories.

7.6 Daily-activity extension (reported separately)

The assignment calls this a carbon audit of daily activity, so the auditor's personal travel and diet were also estimated. They are kept outside the building inventory because they fall outside the Guidelines' building boundary. They correspond to Scope 3 categories in the GHG Protocol and ISO 14064-1.

- **MTR:** 11,820 p-km/yr. This covers an 18 km commute each way, six days a week for 48 weeks, plus three other trips a month of 22 km each way. The factor is MTR's reported rail intensity of 0.0723 kg CO_2e/p-km [17].
- **Flights:** four flights in the last 12 months (Hong Kong-Delhi, Delhi-Hong Kong, Delhi-Goa, Goa-Hong Kong). Each uses its great-circle distance + 8%, with UK DESNZ 2025 economy factors including radiative forcing [18].
- **Diet:** mixed (medium meat-eater), taken from Scarborough et al. at 5.63 kg CO_2e/day [19].

Table 7.6: Auditor's daily-activity footprint

Activity	Quantity	Unit	Factor	kg CO_2e/yr
MTR commute (18 km x 2 x 6 days/week)	10,368	p-km	0.0723 kg CO_2e/p-km	750
Other MTR trips (22 km x 2 x 3/month)	1,452	p-km	0.0723 kg CO_2e/p-km	105
Flights (4 legs, table below)	14,335	p-km	0.1180 kg CO_2e/p-km	1,692
Diet - mixed (medium meat-eater)	365	days	5.6300 kg CO_2e/day	2,055
Lifestyle total				**4,601**

Table 7.7: Flight legs

Leg	Great-circle km	Haul	km incl. 8% uplift	Factor (kg/p-km)	kg CO_2e
Hong Kong (HKG) - Delhi (DEL)	3,747	Long-haul	4,047	0.11704	474
Delhi (DEL) - Hong Kong (HKG)	3,747	Long-haul	4,047	0.11704	474
Delhi (DEL) - Goa (GOI)	1,503	Short-haul	1,623	0.12580	204
Goa (GOI) - Hong Kong (HKG)	4,276	Long-haul	4,618	0.11704	540

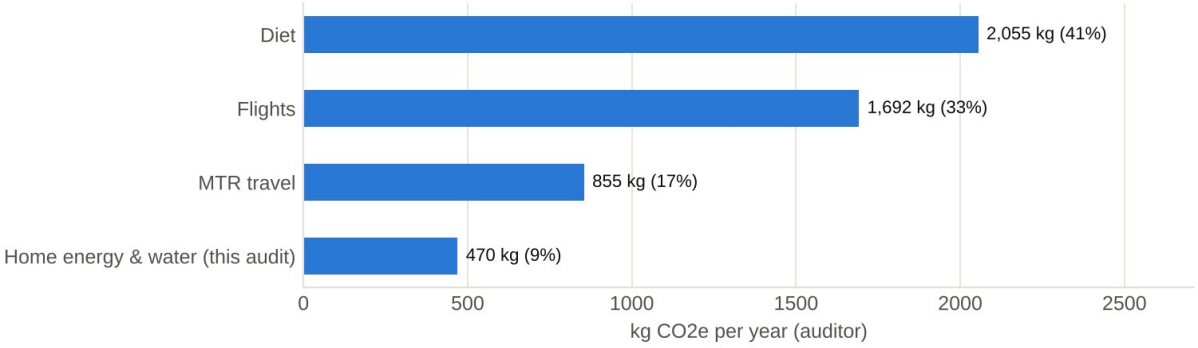


Figure 7.2: The auditor's annual footprint: home energy is a small part

The auditor's total is about **5.1 t CO_2e/yr**, 1.11 × Hong Kong's per-capita figure of 4.58 t [10]. That figure is a territorial inventory, which excludes international aviation and the emissions embodied in imported food, so the comparison is indicative only. The key insight is proportion:

- diet (41%) and flights (33%) outweigh the home's energy and water (9%);
- the same pattern of Scope 3 dominance is typical of corporate footprints.

8 Walk-through findings (EAC §7.4.5, §8.1(i))

The site inspection was structured around the four kinds of obvious saving opportunity listed in EAC §7.4.5:

- (a) maintenance condition;
- (b) excessive provision of temperature, light or flow;
- (c) occupant behaviour;
- (d) match of operating schedules to need.

It also covered the building characteristics listed in §7.3.1(k).

Table 8.1: Findings from the walk-through

No.	Finding	EAC clause	Evidence	Leads to
F1	Dark deposits (dust/mould) on the AC discharge louvres; filter last cleaned 2 months ago	7.4.5(a)	Photo 03b	EMO-03, EMO-16
F2	AC set at 25 °C and run all night (12 h/day in hot months) while the occupants sleep; AC also used 4 h/day in the cool months, including December-February when the mean temperature is 16.5-18.2 °C	7.4.5(b), (d)	Questionnaire Q19-20; HKO normals	EMO-01, EMO-02, EMO-04
F3	Washroom exhaust fan runs 24 h/day regardless of occupancy	7.4.5(d), 7.4.6(a)	Photo 08d, Q35	EMO-11
F4	Kitchen AC (7 kW) left on for 8 h or more; no one is responsible for switching it off	7.4.5(c), (d)	Photo 09f, Q46	EMO-12
F5	Water dispenser keeps 10 L at 98 °C and an ice tank around the clock; SRO already schedules a weekly auto-clean	7.4.6(a)	Photo 09b	EMO-13
F6	Single glazing in a dark aluminium frame facing NW: late-afternoon solar gain in summer	7.3.1(k)	Photos 02a-02c	EMO-03, EMO-20
F7	Lighting is all-LED and washroom lights are on occupancy sensors (good practice), but the kitchen lights are manually switched and burn about 16 h/day	7.4.1(e), 7.4.5(c)	Photos 05a, 08b	EMO-05, EMO-15
F8	AC, refrigerators and kitchen AC are all Grade 1 labelled; operation, not equipment, drives energy use	7.4.4(a)	Photos 03a, 09g, 09i	EMO-19 not viable
F9	24.4 kW instantaneous shower heater: no standing loss but very high peak demand	7.4.4(a)	Photos 08a, 08e	EMO-18
F10	Shower flow and temperature are not displayed; heater dial setting not legible	7.4.5(b)	Photo 08a	EMO-06, -07, -14
F11	Personal routines with easy savings: laundry tumble-dried after every wash, laptops on default power settings for 8 h/day, cooking 3 times a week	7.4.5(c)	Questionnaire F27, I37, J39	EMO-08, -09, -10
F12	No metering of non-AC loads, which limits measurement and verification	7.3.1(h), 7.5.7, 7.6.3	-	EMO-17
F13	Good habits already in place: devices unplugged overnight, paper recycled, filtered tap water instead of bottled water	7.4.5(c)	Questionnaire	Maintain



Photo 09f: kitchen AC (F4)



Photo 09b: dispenser at 98 °C (F5)

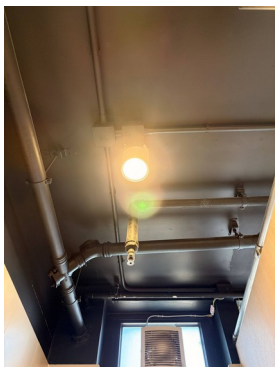


Photo 08c: washroom fan and light (F3)

Figure 8.1: Evidence for findings F3-F5

9 Energy Management Opportunities (EAC §6.2, §7.4-7.5)

21 EMOs were identified and classified using the EAC categories [1]:

- **Category I** (10 EMOs): housekeeping with practically no cost;
- **Category II** (7): operational changes with relatively low cost;
- **Category III** (4): higher capital investment.

They are also typed as in EE-EAes Part 3: LG lighting (2), AC air-conditioning (9), EL electrical (2), LE lift/escalator (0) and others (8).

Savings are priced at HK$1.406/kWh and converted to CO_2e with the CLP 2025 factor. Capital costs are indicative and should be confirmed by quotation. For each EMO with a cost, the cost-benefit analysis gives the simple payback and the life-cycle measures that EAC §7.5.4 asks for. These assume a 4% real discount rate and savings that decline by 1% a year over the service life:

- the net present value (NPV);
- the savings-to-investment ratio (SIR, above 1 = worthwhile);
- the cost of saved energy (CSE), compared with the tariff.

Three EMOs go beyond the usual checklist. EMO-04 recognises that a Hong Kong winter needs no mechanical cooling. EMO-10 replaces the tumble dryer with Hong Kong's traditional air-drying. EMO-15 extends the washroom's sensor control to the kitchen. The register also separates what the occupants can do (Category I) from what only the hall can do, because more than half of the household's energy is used in shared facilities.

Table 9.1: EMO register (household share of savings)

Ref	Cat.	Type	Energy Management Opportunity	kWh/yr	HK$/yr	Cost HK$	Payback (yr)	kg CO_2e/yr
EMO-01	I	AC	Raise the AC set-point from 25 °C to 26 °C and keep the 20 W desk fan running for air movement	24	34	0	Immediate	8
EMO-02	I	AC	Use the AC's built-in timer to switch off 2 h before waking in hot months (12 to 10 h/day)	100	140	0	Immediate	34
EMO-03	I	AC	Clean the AC filter fortnightly in summer, wipe the soiled louvres, close the curtain against late-afternoon NW sun	41	57	0	Immediate	14
EMO-04	I	AC	Cool-season 'fan first': no AC cooling in December-February (HKO mean 16.5-18.2 °C); use the desk fan and open window, Dry mode only on humid days	107	150	0	Immediate	36
EMO-05	I	LG	Use the desk task lights instead of both ceiling battens for 3 of the 6 lit hours	44	62	0	Immediate	15
EMO-06	I	Others	Shorten showers from 5 to 4 minutes (phone timer or a 4-minute song)	127	178	0	Immediate	43
EMO-07	I	Others	Lower the delivered shower temperature from 40 °C to 38 °C (turn the heater dial down rather than mixing cold water)	77	108	0	Immediate	26
EMO-08	I	EL	Laptop power management: lower brightness, sleep after 10 min idle, battery-saver mode	28	40	0	Immediate	10
EMO-09	I	Others	Cook with lids, match pan to the induction zone, batch-cook two meals per session	62	88	0	Immediate	21
EMO-10	I	Others	Air-dry laundry on a rack by the open window instead of tumble-drying (keep the dryer for bulky items in humid spells)	50	70	0	Immediate	17

Ref	Cat.	Type	Energy Management Opportunity	kWh/yr	HK$/yr	Cost HK$	Payback (yr)	kg CO_2e/yr
EMO-11	II	AC	Washroom exhaust fan: replace 24 h running with an occupancy / humidistat run-on control (~8 h/day)	88	123	800	3.2	30
EMO-12	II	AC	Kitchen AC: 7-day timer + occupancy interlock, set-point locked at 25.5 °C (8 to 5 h/day hot, 4 to 2 h/day cool)	33	46	1,000	1.4	11
EMO-13	II	Others	Water dispenser: enable night set-back / timer on hot and ice tanks (00:00-06:00)	9	12	200	1.0	3
EMO-14	II	Others	Fit a WELS Grade-1 low-flow showerhead (9 to 7 L/min) in the shared washroom	141	198	250	0.6	48
EMO-15	II	LG	Kitchen lighting: two ceiling occupancy (PIR) sensors so lights follow use (16 to 8 h/day)	16	23	600	1.6	6
EMO-16	II	AC	Annual professional clean of the window-AC coil and louvres (heavy soiling seen in photo 03b)	24	34	400	11.7	8
EMO-17	II	EL	Buy a plug-in power meter (about HK$120) to measure the laptop, fan and AC and verify savings (EAC §7.6.3)	-	-	120	n/a	-
EMO-18	III	Others	Replace 24.4 kW instantaneous shower heaters with central air-source heat-pump water heating (COP about 3.5) and storage	455	640	20,000	15.6	155
EMO-19	III	AC	Replace the Grade-1 window AC (CSPF 4.66) with a best-in-class split unit (CSPF about 7.5)	307	431	9,000	20.9	104
EMO-20	III	AC	Solar-control film on the NW single glazing (1.55 m^2)	16	23	800	35.1	6
EMO-21	III	Others	Expand rooftop solar PV under the CLP Feed-in Tariff (the village already has 110 panels)	-	-	0	-	-

Table 9.2: Cost-benefit analysis of each EMO (facility level)

Ref	End-use before (kWh/yr)	Saved	Saving kWh/yr	HK$/yr	Cost HK$	Payback yr	Life yr	NPV HK$	SIR	CSE HK$/kWh	kg CO_2e/yr
EMO-01	811	3%	24.3	34	0	0	-	-	-	-	8
EMO-02	597	17%	99.6	140	0	0	-	-	-	-	34
EMO-03	811	5%	40.5	57	0	0	-	-	-	-	14
EMO-04	107	100%	106.7	150	0	0	-	-	-	-	36
EMO-05	88	50%	43.8	62	0	0	-	-	-	-	15
EMO-06	635	20%	126.9	178	0	0	-	-	-	-	43
EMO-07	635	12%	77.2	108	0	0	-	-	-	-	26
EMO-08	190	15%	28.5	40	0	0	-	-	-	-	10
EMO-09	312	20%	62.4	88	0	0	-	-	-	-	21
EMO-10	62	80%	49.9	70	0	0	-	-	-	-	17
EMO-11	263	67%	175.2	246	800	3.2	10	1,117	2.40	0.56	60
EMO-12	1,259	42%	524.0	737	1,000	1.4	10	4,733	5.73	0.24	178
EMO-13	548	25%	136.9	192	200	1.0	10	1,297	7.49	0.18	47
EMO-14	1,269	22%	282.1	397	250	0.6	5	1,482	6.93	0.20	96

Ref	End-use before (kWh/yr)	Saved	Saving kWh/yr	HK$/yr	Cost HK$	Payback yr	Life yr	NPV HK$	SIR	CSE HK$/kWh	kg CO_2e/yr
EMO-15	526	50%	262.8	369	600	1.6	10	2,275	4.79	0.28	89
EMO-16	811	3%	24.3	34	400	11.7	1	-367	0.08	17.10	8
EMO-17	-	-	0.0	0	120	n/a	5	-120	0.00	-	0
EMO-18	1,269	72%	910.4	1,280	20,000	15.6	15	-6,626	0.67	1.98	310
EMO-19	811	38%	306.8	431	9,000	20.9	12	-5,149	0.43	3.13	104
EMO-20	811	2%	16.2	23	800	35.1	10	-623	0.22	6.08	6

Facility level means the whole washroom or kitchen for hall measures (the household's share is in the register above). Category I measures cost nothing, so their NPV equals the value of the savings for as long as the habit is kept.

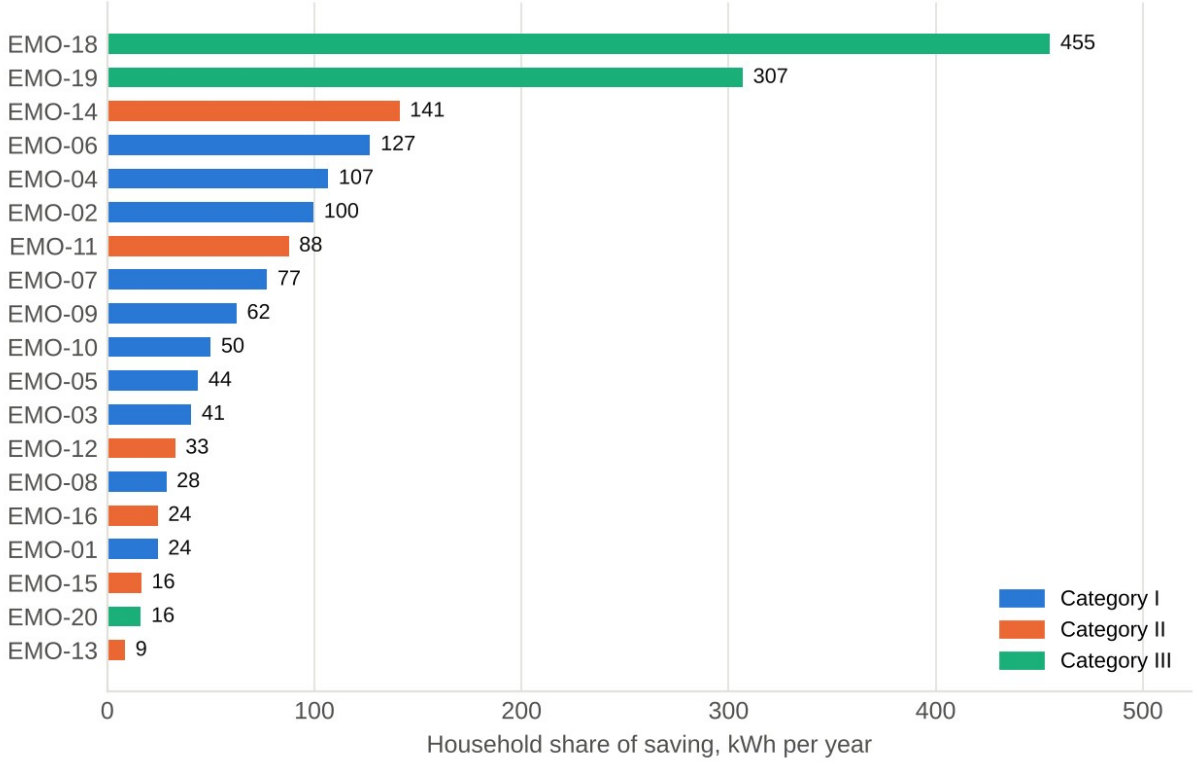


Figure 9.1: Household share of saving by EMO and category

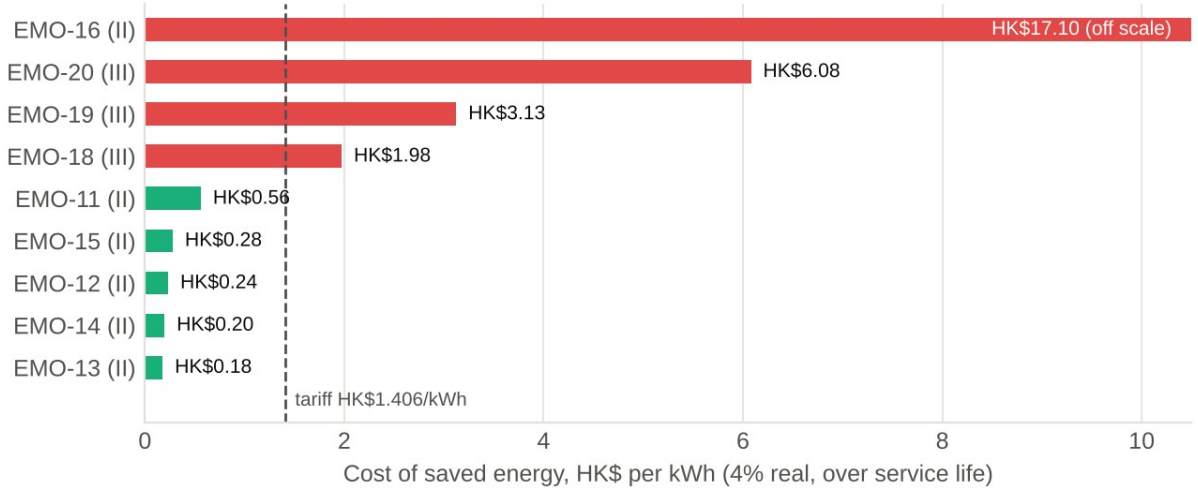


Figure 9.2: Cost of saved energy of the EMOs with a capital cost, against the electricity tariff

9.1 Category I - housekeeping (no cost)

EMO-01 Set-point 26 °C with the desk fan. Each 1 °C rise saves about 3% of AC electricity [14]; the government's guideline for offices is 25.5 °C [2], [14]. With the 20 W desk fan raising air speed, 26 °C stays comfortable (the elevated-air-speed principle of ASHRAE 55). **Saving:** 810.8 kWh × 3%/°C × 1 °C = **24.3 kWh/yr**, 3% of the affected end-use; HK$34/yr and 8 kg CO_2e/yr. **Cost-benefit:** no cost; immediate payback.

EMO-02 AC sleep timer. The AC's built-in timer switches it off 2 h before waking in hot months; the room stays cool from the overnight run and the fan continues. **Saving:** 597.4 kWh (May-Oct) × 2 h ÷ 12 h = **99.6 kWh/yr**, 17% of the affected end-use; HK$140/yr and 34 kg CO_2e/yr. **Cost-benefit:** no cost; immediate payback.

EMO-03 Filter, louvres and curtain. Clean the washable filter every two weeks in the cooling season, as the Guidelines' energy-saving tips advise [2], wipe the soiled louvres and close the curtain against the late-afternoon NW sun. The saving is set at 5%: the low end of the US DOE's 5-15% for replacing a clogged filter [32], consistent with LBNL's finding that coil fouling degrades efficiency by under 5% [26]. **Saving:** 810.8 kWh × 5% = **40.5 kWh/yr**, 5% of the affected end-use; HK$57/yr and 14 kg CO_2e/yr. **Cost-benefit:** no cost; immediate payback.

EMO-04 Cool-season 'fan first'. In December-February the HKO mean temperature is only 16.5-18.2 °C [8], so mechanical cooling is not needed. The 4 h/day of winter AC use reported in the questionnaire can be replaced by the desk fan and the openable window, keeping the AC's Dry mode for humid days only. **Saving:** Dec 35.6 + Jan 35.6 + Feb 35.6 kWh of room-AC energy = **106.7 kWh/yr**, 100% of the affected end-use; HK$150/yr and 36 kg CO_2e/yr. **Cost-benefit:** no cost; immediate payback.

EMO-05 Task lighting. Use the under-shelf desk lights instead of both ceiling battens for 3 of the 6 lit hours. **Saving:** 2 × 20 W × 3 h/day × 365 days ÷ 1000 = **43.8 kWh/yr**, 50% of the affected end-use; HK$62/yr and 15 kg CO_2e/yr. **Cost-benefit:** no cost; immediate payback.

EMO-06 4-minute showers. A phone timer or a 4-minute song shortens each shower by one minute, cutting shower heating energy and water by 20% (about 6.6 m³ of water a year). **Saving:** 634.7 kWh × (1 − 4/5) = **126.9 kWh/yr**, 20% of the affected end-use; HK$178/yr and 43 kg CO_2e/yr. **Cost-benefit:** no cost; immediate payback.

EMO-07 38 °C instead of 40 °C. Turn the heater dial down rather than mixing in cold water, so the heater adds 2 K less to every litre. **Saving:** 634.7 kWh × 2 K ÷ 16.45 K = **77.2 kWh/yr**, 12% of the affected end-use; HK$108/yr and 26 kg CO_2e/yr. **Cost-benefit:** no cost; immediate payback.

EMO-08 Laptop power management. Lower screen brightness, sleep after 10 minutes idle and use battery-saver mode; about 15% of laptop energy. **Saving:** 189.8 kWh × 15% = **28.5 kWh/yr**, 15% of the affected end-use; HK$40/yr and 10 kg CO_2e/yr. **Cost-benefit:** no cost; immediate payback.

EMO-09 Efficient cooking. Use lids, match the pan to the induction zone and batch-cook two meals per session; about 20% of cooking energy. **Saving:** 312.0 kWh × 20% = **62.4 kWh/yr**, 20% of the affected end-use; HK$88/yr and 21 kg CO_2e/yr. **Cost-benefit:** no cost; immediate payback.

EMO-10 Air-dry laundry. Dry most loads on a rack by the open window, with the desk fan in humid spells, and keep the dryer for bulky items. This also saves the dryer's coin/Octopus fee, which is probably higher than the electricity cost used here. **Saving:** 62.4 kWh (2 × 12 min × 3 kW × 52 weeks) × 80% = **49.9 kWh/yr**, 80% of the affected end-use; HK$70/yr and 17 kg CO_2e/yr. **Cost-benefit:** no cost; immediate payback.

The Category I package saves about 660 kWh/yr (25.5% of household electricity) and 224 kg CO_2e/yr at no cost. The AC measures (EMO-01 to EMO-04) cut the occupants' own Octopus spend by about HK$381 a year, 33% of it. The other Category I measures mainly reduce energy paid for by the hall: a classic split incentive.

9.2 Category II - low-cost operational changes

EMO-11 Washroom exhaust-fan control (hall). Replace 24 h running with an occupancy sensor or humidistat with a run-on timer, giving about 8 h a day. Hong Kong's humidity makes run-on control essential to avoid condensation. **Saving:** 30 W × (24 − 8) h × 365 days ÷ 1000 per washroom; household share 2/4 = **175.2 kWh/yr** (household share 87.6 kWh), 67% of the affected end-use; HK$246/yr and 60 kg CO_2e/yr. **Cost-benefit:** HK$800; simple payback 3.2 years; NPV over 10 years HK$1,117; SIR 2.40; cost of saved energy HK$0.56/kWh against the HK$1.406/kWh tariff. **Recommend to hall.**

EMO-12 Kitchen-AC timer and set-point lock (hall). A 7-day timer with an occupancy interlock and a set-point fixed at 25.5 °C, cutting run-hours from 8 to 5 h/day (hot season) and from 4 to 2 h/day (cool season). **Saving:** 0.573 kW × [(8 − 5) h × 184 d + (4 − 2) h × 181 d] per kitchen; household share 2/32 = **524.0 kWh/yr** (household share 32.8 kWh), 42% of the affected end-use; HK$737/yr and 178 kg CO_2e/yr. **Cost-benefit:** HK$1,000; simple payback 1.4 years; NPV over 10 years HK$4,733; SIR 5.73; cost of saved energy HK$0.24/kWh against the HK$1.406/kWh tariff. **Recommend to hall.**

EMO-13 Water-dispenser night set-back (hall). Use the dispenser's timer to switch off the hot and ice tanks from 00:00 to 06:00. The SRO already suspends the machines weekly for auto-cleaning (Photo 09b), so overnight scheduling is feasible. **Saving:** 1.5 kWh/day × 365 × 25% per kitchen; household share 2/32 = **136.9 kWh/yr** (household share 8.6 kWh), 25% of the affected end-use; HK$192/yr and 47 kg CO_2e/yr. **Cost-benefit:** HK$200; simple payback 1.0 years; NPV over 10 years HK$1,297; SIR 7.49; cost of saved energy HK$0.18/kWh against the HK$1.406/kWh tariff. **Recommend to hall.**

EMO-14 WELS Grade-1 showerheads (hall). A WELS Grade-1 head at about 7 L/min instead of the assumed 9 L/min cuts shower energy and water by 22% for all four users of the washroom. **Saving:** 4 users × 317.4 kWh × (1 − 7/9) per washroom; household share 2/4 = **282.1 kWh/yr** (household share 141.0 kWh), 22% of the affected end-use; HK$397/yr and 96 kg CO_2e/yr. **Cost-benefit:** HK$250; simple payback 0.6 years; NPV over 5 years HK$1,482; SIR 6.93; cost of saved energy HK$0.20/kWh against the HK$1.406/kWh tariff. **Recommend to hall.**

EMO-15 Kitchen lighting occupancy sensors (hall). Two ceiling PIR occupancy sensors (about HK$300 each, installed) let the six LED fittings follow use instead of burning about 16 h a day. **Saving:** 6 × 15 W × (16 − 8) h × 365 d ÷ 1000 per kitchen; household share 2/32 = **262.8 kWh/yr** (household share 16.4 kWh), 50% of the affected end-use; HK$369/yr and 89 kg CO_2e/yr. **Cost-benefit:** HK$600; simple payback 1.6 years; NPV over 10 years HK$2,275; SIR 4.79; cost of saved energy HK$0.28/kWh against the HK$1.406/kWh tariff. **Recommend to hall.**

EMO-16 Annual AC coil clean (hall). An annual professional clean of the coil and louvres saves about 3% of AC energy. The energy payback exceeds the one-year service interval, so the measure is justified by hygiene (Finding F1) and equipment life rather than by energy. **Saving:** 810.8 kWh × 3% = **24.3 kWh/yr**, 3% of the affected end-use; HK$34/yr and 8 kg CO_2e/yr. **Cost-benefit:** HK$400; simple payback 11.7 years; NPV over 1 year -HK$367; SIR 0.08; cost of saved energy HK$17.10/kWh against the HK$1.406/kWh tariff. **Recommend on hygiene grounds; bundle with routine maintenance (EAC §7.6.4).**

EMO-17 Plug-in power meter (occupants). About HK$120, within the stated HK$200 budget. It measures the laptop, fan and AC directly, turning assumed values into measured ones and providing M&V (EAC §7.5.7, §7.6.3). No saving is claimed. **Cost-benefit:** HK$120; no energy payback; NPV over 5 years -HK$120; SIR 0.00. **Implement now (within HK$200 budget).**

9.3 Category III - capital measures and cost-benefit with service life (EAC §7.5.4)

EMO-18 Heat-pump water heating (CityU). A central air-source heat pump (COP about 3.5) with storage would replace the 24.4 kW instantaneous heaters, saving about 72% of shower electricity and greatly reducing the hall's maximum demand. It needs a feasibility study covering demand-charge savings and the space and structural limits of the MiC modules. **Saving:** 4 users × 317.4 kWh × (1 − 0.99/3.5) per washroom; household share 2/4 = **910.4 kWh/yr** (household share 455.2 kWh), 72% of the affected end-use; HK$1,280/yr and 310 kg CO_2e/yr. **Cost-benefit:** HK$20,000; simple payback 15.6 years; NPV over 15 years

-HK$6,626; SIR 0.67; cost of saved energy HK$1.98/kWh against the HK$1.406/kWh tariff. **Further study (also cuts peak demand).**

EMO-19 Replace the window AC (CityU). A best-in-class split unit (CSPF about 7.5; Grade-1 units range from 5.06 to 7.51 [27]) would replace a unit that is new and already Grade 1. **Saving:** 810.8 kWh × (1 − 4.66/7.5) = **306.8 kWh/yr**, 38% of the affected end-use; HK$431/yr and 104 kg CO_2e/yr. **Cost-benefit:** HK$9,000; simple payback 20.9 years; NPV over 12 years -HK$5,149; SIR 0.43; cost of saved energy HK$3.13/kWh against the HK$1.406/kWh tariff. **Not viable - payback exceeds service life.**

EMO-20 Solar-control window film (CityU). Film on only 1.55 m² of NW glazing saves about 2% of AC energy. **Saving:** 810.8 kWh × 2% = **16.2 kWh/yr**, 2% of the affected end-use; HK$23/yr and 6 kg CO_2e/yr. **Cost-benefit:** HK$800; simple payback 35.1 years; NPV over 10 years -HK$623; SIR 0.22; cost of saved energy HK$6.08/kWh against the HK$1.406/kWh tariff. **Not viable at present.**

EMO-21 Rooftop PV (CityU). A viability statement under EAC §7.4.4(d) and §8.1(l)(iii); see Section 9.4.

The NPVs above already include a 1%/yr decline in savings (EAC §7.5.4(b)). Every Category III measure has a cost of saved energy above the tariff, so none is cost-effective on energy grounds alone. EMO-18 deserves study because it also cuts the hall's peak demand.

9.4 Energy recovery and renewable energy (EAC §7.4.4(c)-(d), §8.1(l)(iii))

Energy recovery: drain-water heat recovery could pre-heat shower water. However, with one shower at a time, instantaneous heating and the tight MiC washroom pods, the recoverable heat is small and the retrofit intrusive, so it is not recommended.

Renewable energy: the village already has 110 rooftop PV panels [15]. Additional roof or façade PV would earn CLP's Feed-in Tariff: HK$3/kWh for 10-200 kW and HK$2.5/kWh for 200 kW-1 MW [22]. The economics are attractive at hall scale but outside the household's control, so it is recommended for study by CityU (EMO-21).

Table 9.3: Combined impact by category (household share)

Category	No. of EMOs	kWh/yr saved	% of household electricity	kg CO_2e/yr (CLP)	Facility-level kWh/yr
Category I	10	660	25.5%	224	660
Category II	7	311	12.0%	106	1,405
Category III	4	778	30.1%	265	1,233
All EMOs*	**21**	**1,749**	**67.6%**	**595**	**3,299**

**Savings interact (e.g. shorter showers and low-flow heads both reduce the same hot water), so totals are indicative rather than strictly additive. The Category III total includes measures found not viable.*

10 Scaling up to Sha Tin District

Ma On Shan lies in **Sha Tin District**, the most populous of Hong Kong's 18 districts. This chapter turns the audit into two scaling factors, energy per capita and energy per square metre, and applies them to the district's population and domestic floor area. The population and household data come from the Census and Statistics Department [33]; the floor-area data come from the Housing Department and Census records [23], [34]-[37]. The results are compared with each other (i) and a composite factor linking energy, floor area and residents is developed (ii). Electricity is the quantity scaled, because it is what this all-electric household uses.

10.1 Factors from the audit

Table 10.1: Scaling factors derived from the audited household

Factor	Value	Unit	Basis
Energy per capita, e_pc	1,293	kWh/person/yr	Household electricity 2,586 kWh ÷ 2 residents
Energy per square metre, e_m2	172.4	kWh/m²/yr	Household electricity ÷ 15 m² dwelling floor area (shared-facility energy included)
For reference: room EUI	81.3	kWh/m²/yr	Energy used inside the room only (EAC EUI basis)
Floor area per person in the audit, s	7.5	m²/person	Half the Hong Kong median

10.2 Sha Tin District data

The district's domestic floor area is estimated in two independent ways:

- **Method 1** multiplies the population by the 2021 Census median floor area per person (16.0 m²) [23].
- **Method 2** builds the floor area up by housing type. Public rental housing uses the Housing Authority's household count for Sha Tin [35] and its average living space of 14.1 m² per person at 2.6 persons per household [34]. Subsidised-sale and private flats use the 2016 By-census median floor areas (35 and 47 m²) [36], split in the 2021 Census proportions [37].

The two estimates agree within 8%.

Table 10.2: Sha Tin District population, households and domestic floor area

Quantity	Value	Unit	Source
Population (land-based non-institutional), 2025	687,300	persons	C&SD [33]
Domestic households, 2025	248,200	households	C&SD [33]
Average household size	2.7	persons	C&SD [33]
Method 1: domestic floor area (population × 16.0 m²)	10,996,800	m²	Census median [23]
Public rental households (Housing Authority) × 14.1 m² × 2.6 persons	77,100 → 2,826,486 m²		Housing Department [34], [35]
Subsidised-sale households (37.2% of the rest) × 35 m²	2,228,810	m²	[36], [37]
Private households × 47 m²	5,048,727	m²	[36], [37]
Method 2: domestic floor area by housing type	10,104,023	m²	-8.1% vs method 1
Floor area per person (method 1)	16.0	m²/person	

10.3 District results and comparison (i)

Table 10.3: Sha Tin residential electricity by scaling method

Method	Factor	Scaling quantity	GWh/yr	vs EMSD-based benchmark
(1) Per capita: e_pc x population	1,293 kWh/person/yr	687,300 persons	889	-27%
(2) Per m²: e_m2 x floor area (method 1)	172 kWh/m²/yr	10,996,800 m²	1,896	+55%
(2b) Per m²: e_m2 x floor area (method 2)	172 kWh/m²/yr	10,104,023 m²	1,742	+43%
(3) Per household: household electricity x households	2,586 kWh/household/yr	248,200 households	642	-47%
(4) Composite: a x floor area + b x residents (method 1 area)	62 a, kWh/m²/yr	10,996,800 m²	1,250	+2%
(4b) Composite with method 2 floor area	62 a, kWh/m²/yr	10,104,023 m²	1,195	-2%
(4c) Composite, allowing for gas water heating and cooking	473 kWh/person/yr moved to gas	343,650 residents using gas	1,087	-11%
Benchmark A: HK residential electricity per person (EMSD 2021) x population	1,778 kWh/person/yr	687,300 persons	1,222	-
Benchmark B: CLP residential sales per account (2025) x households	3,941 kWh/account/yr	248,200 households	978	-20%

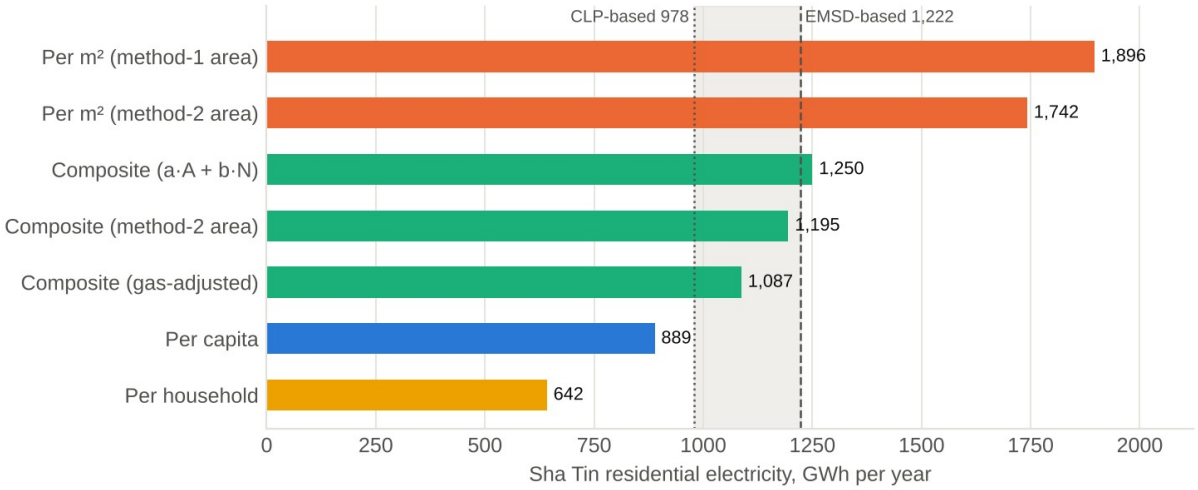


Figure 10.1: Sha Tin residential electricity estimated by each method, against two top-down benchmarks

Per capita: 1,293 kWh × 687,300 residents = **889 GWh/yr**. **Per m²:** 172.4 kWh × 10,996,800 m² = **1,896 GWh/yr**. The two differ by a factor of **2.13**, and this ratio is exactly the district's floor area per person divided by the audited household's (16.0 ÷ 7.5 m² = 2.13). The two factors would agree only if Sha Tin residents had as little space as a student in a twin room.

- The **per-capita** method **under**-estimates (-27% against the EMSD-based benchmark). A student with 7.5 m² cools and lights far less space than a typical resident with 16 m².
- The **per-m²** method **over**-estimates (+55%). About two-thirds of the household's electricity (hot water, cooking, devices, laundry and shared services) depends on the number of people, not on floor area. A dense dormitory packs that energy into very few square metres.
- The **per-household** method (642 GWh/yr) is lowest, because the audited household has 2 people and 15 m², against 2.7 people and about 44 m² for the district.

Two top-down benchmarks bracket the true value. Hong Kong's residential electricity per person (EMSD 2021) × Sha Tin's population gives **1,222 GWh/yr** [38]. CLP's residential sales per account (2025) × Sha

Tin's households gives **978 GWh/yr** [39]. No official district-level electricity figure is published, so these are the best available checks.

10.4 A composite factor for energy, floor area and residents (ii)

The audit's end-uses split naturally into two groups:

- **area-driven** loads that scale with conditioned and lit floor area: room AC and room lighting, 928 kWh/yr (36%);
- **resident-driven** loads that scale with the number of people: hot water, cooking, laptops and phones, the fan, laundry and the shared services, 1,659 kWh/yr.

Dividing each by its driver gives the composite household factor:

E = a × A + b × N, with **a = 61.8 kWh per m² per year** and **b = 829 kWh per resident per year** (A = floor area, N = residents).

Per person: **e(s) = b + a × s**, where s is floor area per person. Per m²: **e(d) = a + b × d**, where d is residents per m².

The single-variable factors are special cases. The per-capita factor holds only at s = 7.5 m²; the per-m² factor holds only at d = 0.13 persons/m². The composite reproduces the audited household exactly.

Table 10.4: The composite factor applied to typical Sha Tin households

Household type	Floor area per person s (m²)	Residents N	Floor area A (m²)	E = aA + bN (kWh/yr)	kWh per person	kWh per m²
Audited student room (this audit)	7.5	2.0	15.0	2,586	1,293	172.4
Sha Tin public rental flat	14.1	2.6	36.7	4,423	1,701	120.7
Sha Tin average household	16.0	2.7	43.2	4,911	1,819	113.7
Private flat (47 m², Sha Tin household size)	17.4	2.7	47.0	5,146	1,906	109.5

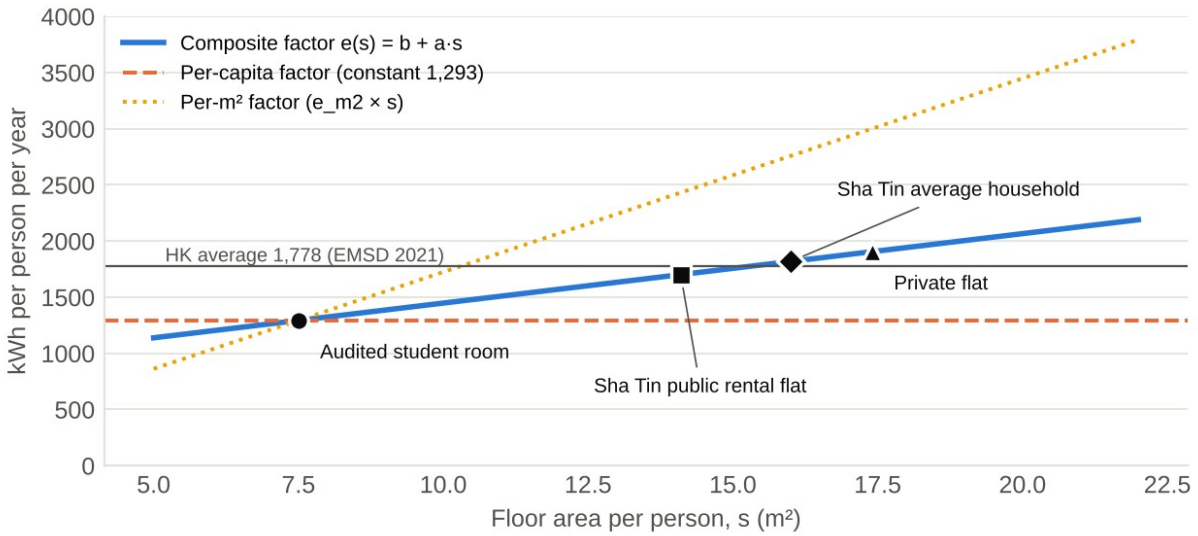


Figure 10.2: Electricity per person against floor area per person - composite, per-capita and per-m² factors

Applied to Sha Tin, the composite factor gives 61.8 × 10,996,800 m² + 829 × 687,300 residents = **1,250 GWh/yr**. That is within 2% of the EMSD-based benchmark, and 1,195 GWh/yr with the method-2 floor area. For a Sha Tin household of average size (2.7 persons, 43.2 m²) it predicts 4,911 kWh/yr. That is within 0.4% of Hong Kong's 2021 residential average of 4,930 kWh per household [38]. Agreement this close is partly fortunate, but it shows that a factor using both drivers transfers between dwelling types far better than either single factor.

Limitations of the scale-up.

- The factors come from one all-electric student household. Many Hong Kong homes heat water and cook with Towngas or LPG. If half of Sha Tin's residents do, the composite electricity estimate falls to 1,087 GWh/yr (-11%).
- AC use per m² in a small, fully occupied room may differ from that in larger flats with unoccupied rooms.
- The district data mix years (2016 By-census floor-area medians, 2021 Census housing mix, 2025 population and 2026 Housing Authority counts) and use different area definitions (internal floor area for public rental housing, saleable area for private flats).
- The benchmarks are territory-wide averages, not Sha Tin measurements.

11 Recommendations, implementation and M&V (EAC §7.6)

11.1 Implementation plan

Table 11.1: Action plan

When	Action	Who	Cost	Indicator to track
Now	EMO-01 to EMO-10: set-point 26 °C + fan, no AC cooling in Dec-Feb, timer-off, filter cleaning, task lighting, 4-min showers at 38 °C, laptop settings, efficient cooking, air-dried laundry	Both occupants	HK$0	Octopus spend per month; shower timer; dryer uses
Within 1 month	EMO-17: buy a plug-in power meter; measure laptop, fan and AC for 2 weeks	Auditor	About HK$120	Measured W and kWh
Within 1 month	Send the findings memo (Section 11.3) to the Student Residence Office	Auditor	HK$0	Reply / action from SRO
Next maintenance cycle	EMO-11 to EMO-16: exhaust-fan control, kitchen-AC timer, dispenser set-back, Grade-1 showerheads, kitchen sensors, AC coil clean	SRO / hall maintenance	HK$200-1,000 each	Fan, light and AC run-hours; kitchen meter
Next capital plan	EMO-18 and EMO-21: heat-pump hot water and PV feasibility studies	CityU Campus Development	Study cost	Business case

11.2 Measurement and verification plan (EAC §7.5.3, §7.6.3)

Table 11.2: Metering points, devices and intervals

Quantity	Metering point / device	Interval and duration	Adjustment
Room AC energy	Octopus AC balance (screenshot weekly); plug-in meter if the socket allows	Weekly, 4 weeks before and 4 weeks after the EMOs, in both seasons	Normalise by cooling degree-days from HKO daily data
Laptop and fan energy	Plug-in power meter	2 weeks each	Per hour of use
Shower flow and time	Bucket test (L/min) and phone timer	Once; then a weekly sample	Inlet temperature by month (HKO)
Kitchen AC, lights and exhaust fan hours	Run-hour log or smart-plug data (hall)	Continuous for 1 month	Occupancy (users per day)
Baseline record	This workbook (Inputs, AC_Baseline, Monthly)	Kept with the report	Re-run with measured values

11.3 Communication with hall management (EAC §7.4.3)

More than half of the household's energy is used in shared facilities, so the most effective EMOs depend on the Student Residence Office. A short memo should be sent to the SRO. It should:

- summarise findings F1, F3-F5, F7, F9 and F10;
- quantify the facility-level savings of EMO-11 to EMO-15 (about 1,381 kWh/yr per washroom and kitchen, NPV positive for each) and recommend the coil clean (EMO-16) on hygiene grounds;
- suggest piloting them on one floor, with before-and-after readings from the building management system.

This mirrors a corporate audit, where landlord-controlled central services are reported to the responsible persons for follow-up.

11.4 Further studies (EAC §7.6.5)

- Obtain CityU's actual Octopus AC tariff (HK$/kWh) to replace the CLP-average proxy. This is the largest single uncertainty.

- Measure the exhaust-fan, desk-light, ceiling-batten and kitchen-light power ratings, and the washroom and kitchen floor areas.
- Record the water heater's dial setting, and measure the shower flow with a bucket test.
- At hall level: sub-meter the washroom heaters and kitchens, and study the demand-charge benefits of heat-pump water heating.
- At district level: refine the composite factor with metered data from flats of different sizes, and with households that use gas.

12 Uncertainty, limitations and sensitivity

Each key assumption was varied one at a time, and household Scope 2 emissions were recalculated in the workbook (table and figure below).

The territory-wide default factor of 0.7 kg/kWh raises the result by 106%. This is a reporting convention rather than an uncertainty, which is why both factors are reported. Among the physical assumptions, the largest effects are:

- **AC energy:** -12% to +8% across the alternative Octopus tariffs and the occupants' flat HK$120 estimate;
- **roommate's habits:** -12% if the roommate's personal use is half the auditor's;
- **shower flow rate:** -5% to +8% for 7-12 L/min.

Combining the main independent uncertainties by root-sum-square gives about **±20%** on household electricity, i.e. roughly 2,070-3,103 kWh/yr. The district factors inherit this uncertainty; the larger uncertainty there is how representative one dormitory household is (Section 10.4).

Table 12.1: One-at-a-time sensitivity (household, CLP factor unless stated)

#	Scenario	Value	Room AC kWh	Household kWh	Scope 2 kg	Change
0	Base case	CLP 0.34; HK$1.406/kWh	811	2,586	879	+0.0%
1	CLP 2024 emission factor	0.380	811	2,586	983	+11.8%
2	Territory-wide default factor	0.700	811	2,586	1,810	+105.9%
3	Octopus AC rate HK$1.20/kWh (CUHK level)	1.20	950	2,726	927	+5.4%
4	Octopus AC rate HK$2.31/kWh (HKU level)	2.31	494	2,269	771	-12.3%
5	Stated flat average HK$120/month	120.00	1,024	2,800	952	+8.2%
6	Bottom-up AC (label 362 kWh/1,200 h)	362.00	884	2,660	904	+2.8%
7	Shower flow 7 L/min (Grade-1 head)	7.00	811	2,445	831	-5.5%
8	Shower flow 12 L/min	12.00	811	2,798	951	+8.2%
9	Laptop average draw 30% of adapter	0.300	811	2,510	854	-2.9%
10	Laptop average draw 80% of adapter	0.800	811	2,700	918	+4.4%
11	Roommate uses half the auditor's personal loads	0.500	811	2,275	774	-12.0%
12	Kitchen allocated over 20 active users (not 32)	20.00	811	2,694	916	+4.1%
13	One month away (11 of 12 months occupied)	0.917	743	2,398	815	-7.3%

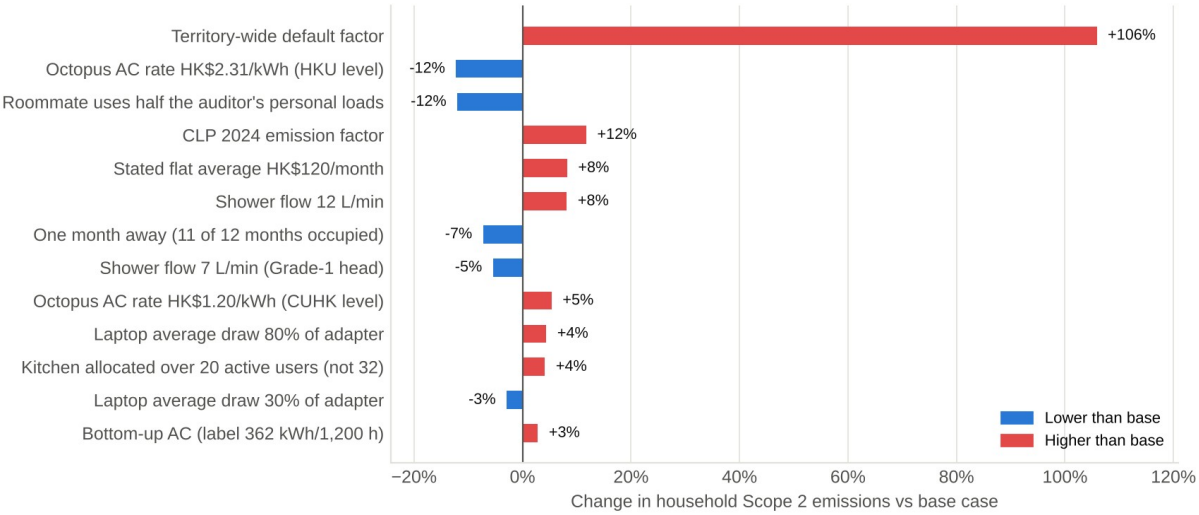


Figure 12.1: Sensitivity of household Scope 2 emissions to each assumption

12.1 Limitations

- No sub-metering: only the AC is metered (through spend), and all other loads are estimated from ratings and reported hours.
- 33 of 124 inputs are typical assumed values. Among them, lighting and fan wattages are assumed because no labels were visible.
- The roommate's personal loads are assumed equal to the auditor's, and shared loads are allocated per user, which ignores differences in individual use.
- The label-based AC estimate uses the HK test-condition average, not this room's measured load; cold-water inlet temperature is taken as the monthly mean air temperature.
- EMO costs are indicative, and life-cycle results depend on the assumed 4% discount rate and 1%/yr degradation.
- The Guidelines (2010) are dated; current factors were substituted, using the Guidelines' methods. Daily-activity factors (DESNZ, Scarborough) are not Hong Kong-specific.
- District data were taken from official publications as reported in search extracts of those publications; the figures should be checked against the source tables before reuse.

13 Conclusions and lessons for the corporate carbon audit

13.1 Conclusions

- The household uses about **2,586 kWh/yr** of electricity: 1,293 kWh per person, and a room EUI of 292.7 MJ/m²/yr. This emits **939 kg CO_2e/yr** on CLP's 2025 factor, or 1,870 kg on the 0.7 default. Since move-in (January-September 2026) it has used about 1,956 kWh.
- Air-conditioning (34.4%) and shower hot water (24.5%) dominate. Their seasonality runs in opposite directions, which flattens the monthly profile.
- The equipment is already efficient (Grade-1 labels, LED lighting). The savings lie in operation: set-points, schedules, run-hours of shared plant, and shower and laundry habits.
- 52.8% of the household's energy is used in shared facilities, so the hall management is an essential partner.
- No-cost Category I measures save about 25.5%. The five main Category II measures at hall level (EMO-11 to EMO-15) pay back in 0.6-3.2 years with positive NPV. Replacing the AC is not viable.
- Scaled to Sha Tin, per-capita and per-m² factors give 889 and 1,896 GWh/yr. A composite factor (a = 61.8 kWh/m², b = 829 kWh/person) gives 1,250 GWh/yr, close to the EMSD-based 1,222 GWh/yr.

13.2 Lessons for the corporate carbon audit

1. **Set the boundary deliberately.** In a hostel, as in a multi-tenant office, the energy a user causes sits partly in landlord-controlled central services. Organisational and operational boundaries, and the allocation rules between them, must be decided and documented before any data is collected.
2. **Build a data hierarchy and fill gaps transparently.** Metered data, then nameplate ratings, then reported behaviour, then published values, then assumptions. Each value is tagged and stress-tested.
3. **Triangulate.** Two independent estimates (financial and engineering) that agree within about 10% give far more confidence than one.
4. **Report emission factors explicitly.** Hong Kong's Guidelines require both the territory default and the utility-specific factor; the choice alone changes the answer by a factor of two.
5. **Normalise with more than one driver.** Energy per m² or per person alone misleads when occupancy density differs; a two-driver factor transfers between sites far better, as corporate benchmarking also finds.
6. **Expect Scope 3 to dominate.** Travel and food far exceeded the home's energy, as purchased goods and business travel typically do in corporate footprints.
7. **Behaviour and governance beat hardware.** Split incentives (who pays versus who decides) shaped the EMO list as much as engineering did.
8. **Plan M&V from day one.** Without metering points, savings cannot be verified (EAC §7.5-7.6).

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Appendix A - Questionnaire responses (summary)

The full answers are in audit-inputs/QUESTIONNAIRE.md in the project repository.

Question	Response
A1-A5 Admin	Jayash Sahu; SEE8115; Hall 13, 10/F, Room 1003B; survey 01/01/2026 to present; roommate not named
B6 Floor area	About 15 m²
B7 Window direction	328° NW (phone compass)
B8 Window	2 × 0.67 m² + 0.209 m²; single glazing; metal frame; curtains
B9 Direct sun	Afternoon, about 40 minutes
B10 Building age	Answered 6-12-2024; the residence opened in 2024 [15]
B11-B12 Washroom / kitchen	Washroom size unknown (4 users); kitchen shared by 32 residents
C13-C15 Occupancy	Weekdays 22:00-11:00; weekends in hall (Sat 20 h, Sun 18 h); 1-month break in the last 10 months
D16 AC	Window, Midea 1.5 HP R32 inverter MW-12CRF8E, 11,400 Btu/h, input 1,192 W
D17-D18 AC charging	Octopus card, rate unknown; about HK$120/month combined; hot months HK$140, cold months HK$50
D19-D21 AC use	25 °C, on all night; summer 12 h/day, winter 4 h/day; never with window open; filter cleaned 2 months ago
D22 Fan	One 20 W electric fan, 12 h/day
E23-E26 Lighting	2 LED ceiling lights (6 h/day); 2 LED desk lights (wattage unknown); washroom sensor lights (about 3 h)
F27-F31 Electronics	65 W laptop charger, 8 h/day; no monitor/PC/console; phone 1 h/day; no router/printer; nothing left on standby
G32 Other appliances	None
H33-H35 Washroom	Instantaneous tankless heater, 380 V (Stiebel Eltron DHB-E 27); 5-min showers; exhaust fan always on
I36-I38 Kitchen	2 Panasonic NR-BV361B fridges (263 kWh/yr); Toshiba MW2-MM24PC microwave (800 W output); 2 Philips HD4911/80 induction (2,100 W); Aquatek AQ-1323T dispenser (750 W/160 W); cooking 3×/week, 1 h; about 20 active users
J39 Laundry	1 load/week, 4 kg, 45 min wash, 12 min drying
K40-K41 Water and paper	Filtered water 2 L/day; 50 pages printed + 50 sheets used per semester; paper recycled
L42-L45 Daily activity	MTR 18 km × 2 × 6/week; 3 trips/month of 22 km × 2; canteen 7 + self-cooked 7 meals/week, mixed diet; 4 flights: HK-Delhi, Delhi-HK, Delhi-Goa, Goa-HK
M46-M48 Observations	Kitchen AC on 8 h+ (Toshiba RAS-24J2KCV-HK, 688 kWh/1,200 h, 7.00 kW, R410A); no changes allowed in the room; budget HK$200

Appendix B - Photo log

26 photographs were taken during the audit and are stored in the photos/ folder. Each was used to read a rating or label, or to evidence a finding.

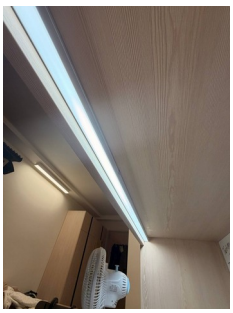
Ref	File	What it shows	Data extracted
01	01_room_overview.jpg	Room 1003B: twin beds, desks, ceiling LED battens, window AC above NW window	Room layout; equipment present
02a	02a_window_view.jpg	Window and outlook over Tolo Harbour / Ma On Shan	Single glazing, dark aluminium frame, fly screen
02b	02b_window_fixed_pane_measured.jpg	Bottom fixed pane measured 1.10 m x 0.17-0.19 m	Glazing area 0.209 m²
02c	02c_window_casement_measured.jpg	Casement measured 0.53 m x 1.26 m = 0.67 m²	Two casements 1.34 m²
02d	02d_compass_328NW.jpg	Compass 328° NW, 22°26'0"N 114°14'32"E, Ma On Shan, 44 m	Orientation and location
02e	02e_curtain_closed.jpg	Full-height grey curtain closed; AC at upper left	Internal shading
03a	03a_ac_specification.jpg	Midea MW-12CRF8E specification / energy label data	3.34 kW, 1,192 W, R32, Grade 1, 362 kWh, CSPF 4.66
03b	03b_ac_front_louvres.jpg	AC front: dark deposits on discharge louvres; Eco, Timer, Dry modes	Maintenance finding; controls available
05a	05a_ceiling_led_battens.jpg	Two linear LED ceiling battens	Lighting inventory
05b	05b_desk_led_and_fan.jpg	Under-shelf LED desk light and desk fan	Task lighting
06a	06a_laptop_adapter_65W.jpg	HP 65 W adapter TPN-CA17, Efficiency Level VI	Laptop power
06b	06b_desk_fan_label_20W.jpg	Fan label: ZW02-180, 180 mm, 20 W, 220 V	Fan power
08a	08a_water_heater_DHB-E.jpg	Stiebel Eltron DHB-E electronic instantaneous heater, 30-60 °C dial	Hot-water system type
08b	08b_washroom_downlights.jpg	Two LED downlights, exposed services, sensor	Washroom lighting
08c	08c_washroom_light_and_fan.jpg	Downlight and window exhaust fan	Washroom lighting / ventilation
08d	08d_washroom_exhaust_fan.jpg	J.E.E. window exhaust fan (runs 24 h)	Ventilation finding
08e	08e_water_heater_spec.jpg	Heater specification: 380 V, max 60 °C	Electrical supply
09a	09a_dispenser_spec.jpg	Aquatek AQ-1323T: 750 W heater, 160 W compressor, 10 L hot / 7.9 L cold / 3 L ice	Dispenser ratings
09b	09b_water_dispenser.jpg	Dispenser in kitchen, hot water at 98 °C; SRO cleaning notice	Standing losses
09c	09c_induction_cooker_spec.jpg	Philips HD4911: 2,100 W, EEL Grade 2	Cooking power
09d	09d_induction_cooker_product.jpg	Philips 5000 Series HD4911/80 (retailer image)	Model identification
09e	09e_induction_energy_label.jpg	Energy label Grade 2, thermal efficiency 88.1% (I220011/22)	Cooking efficiency
09f	09f_kitchen_split_ac.jpg	Toshiba split AC in kitchen (also used as photo 12 - left on 8 h+)	Kitchen AC
09g	09g_kitchen_ac_energy_label.jpg	Label: 688 kWh/yr @1,200 h, 7.00 kW, R410A, Grade 1	Kitchen AC energy
09h	09h_microwave.jpg	Toshiba MW2-MM24PC 24 L, 800 W output	Kitchen inventory
09i	09i_fridge_energy_label.jpg	Panasonic NR-BV361B: Grade 1, 263 kWh/yr, 233 + 99 L	Refrigeration energy



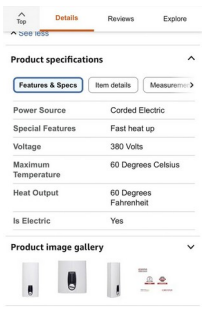
02b: fixed pane 1.10 × 0.19 m



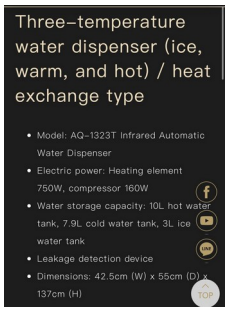
02e: curtain closed



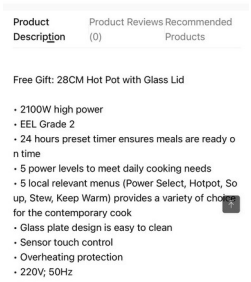
05b: desk LED and fan



08e: heater spec, 380 V



09a: dispenser ratings



09c: hob 2,100 W, Grade 2



09d: Philips HD4911/80



09h: microwave 800 W output

Figure B.1: Additional evidence photographs

Appendix C - Input data register

All 124 inputs used in the calculations, with unit, data-quality class and source ([n] = reference number). Values marked "Assumed" are typical values and are highlighted in yellow in the workbook.

Parameter	Value	Unit	Quality	Src
Occupants of Room 1003B (household)	2	persons	Reported	[29]
Users of the shared washroom	4	persons	Reported	[29]
Residents sharing the floor kitchen (allocation basis)	32	persons	Reported	[29]
Hours in room per weekday (22:00-11:00)	13	h/day	Reported	[29]
Hours in room on Saturday	20	h/day	Reported	[29]
Hours in room on Sunday	18	h/day	Reported	[29]
Internal floor area of Room 1003B	15	m2	Reported	[29]
Days in hot season (May-Oct)	184	days	Derived	
Days in cool season (Nov-Apr)	181	days	Derived	
Energy conversion	3.6	MJ/kWh	Published	[3]
Specific heat of water	4.186	kJ/kg.K	Published	
Electricity price used to convert AC spend to kWh (proxy)	1.406	HK$/kWh	Published	[6]
Comparator hostel AC rate - low	1.2	HK$/kWh	Published	[25]
Comparator hostel AC rate - high	2.31	HK$/kWh	Published	[25]
CLP Power HK emission factor (2025, latest)	0.34	kg CO_2e/kWh	Published	[5]
CLP Power HK emission factor (2024)	0.38	kg CO_2e/kWh	Published	[5]
Territory-wide default electricity factor	0.7	kg CO_2e/kWh	Published	[2]
WSD unit electricity - fresh water	0.624	kWh/m3	Published	[11]
WSD unit electricity - salt (flushing) water	0.362	kWh/m3	Published	[11]
DSD unit electricity - sewage treated	0.29	kWh/m3	Published	[12]
Fraction of fresh water entering sewer (residential)	1	-	Published	[2]
Paper to landfill emission factor	4.8	kg CO_2e/kg	Published	[2]
GWP of HFC-32 (R32)	650	-	Published	[2]
GWP of R-410A	1725	-	Published	[2]
Annual refrigerant leak rate (screening)	0.04	fraction/yr	Published	[21]
Octopus AC spend - hot month (May-Oct), both occupants	140	HK$/month	Reported	[29]
Octopus AC spend - cool month (Nov-Apr), both occupants	50	HK$/month	Reported	[29]
Stated overall average AC spend	120	HK$/month	Reported	[29]
Rated input power (cooling)	1.192	kW	Nameplate / label	[28]
Rated cooling capacity	3.34	kW	Nameplate / label	[28]
Cooling seasonal performance factor	4.6618	-	Nameplate / label	[28]
Label annual energy consumption (cooling)	362	kWh/yr	Nameplate / label	[28]
Operating hours behind HK label figure	1200	h/yr	Published	[9]
AC operating hours - typical hot-season day	12	h/day	Reported	[29]
AC operating hours - typical cool-season day	4	h/day	Reported	[29]
Usual set-point	25	degC	Reported	[29]
Refrigerant charge (R32) - screening value	0.4	kg	Assumed	

Parameter	Value	Unit	Quality	Src
AC energy saving per +1 degC set-point	0.03	fraction	Published	[14]
Ceiling LED linear battens	2	no.	Nameplate / label	[29]
Power per ceiling batten	20	W	Assumed	
Ceiling lights on	6	h/day	Reported	[29]
Under-shelf LED desk lights	2	no.	Reported	[29]
Power per desk light	10	W	Assumed	
Desk lights on	4	h/day	Assumed	
Laptop adapter rating	65	W	Nameplate / label	[29]
Average laptop draw as fraction of adapter rating	0.5	-	Assumed	
Laptop use	8	h/day	Reported	[29]
Laptops in room (roommate assumed same as auditor)	2	no.	Assumed	
Phone charger power	20	W	Assumed	
Phone charging	1	h/day	Reported	[29]
Phones charged (both occupants)	2	no.	Assumed	
Desk fan rated power	20	W	Nameplate / label	[29]
Desk fan use	12	h/day	Reported	[29]
Showers per person per day	1	no./day	Reported	[29]
Shower duration	5	min	Reported	[29]
Shower flow rate	9	L/min	Assumed	
Delivered shower temperature	40	degC	Assumed	
Instantaneous heater efficiency	0.99	-	Assumed	[16]
Water heater rating at 380 V	24.4	kW	Published	[16]
Washroom exhaust fan power	30	W	Assumed	
Exhaust fan running	24	h/day	Reported	[29]
LED downlights	3	no.	Nameplate / label	[29]
Power per downlight	10	W	Assumed	
Sensor lights on	3	h/day	Reported	[29]
Refrigerators	2	no.	Reported	[29]
Label consumption per refrigerator	263	kWh/yr	Nameplate / label	[29]
Water dispenser standby + dispensing energy	1.5	kWh/day	Assumed	[20]
Kitchen split AC label consumption	688	kWh/yr	Nameplate / label	[29]
Kitchen AC running - hot season	8	h/day	Reported	[29]
Kitchen AC running - cool season	4	h/day	Assumed	
Kitchen AC refrigerant charge (R410A) - screening	1.4	kg	Assumed	
Kitchen light fittings	6	no.	Assumed	
Power per kitchen fitting	15	W	Assumed	
Kitchen lights on	16	h/day	Assumed	
Cooking sessions per person	3	per week	Reported	[29]
Induction hob on per session	1	h	Reported	[29]
Average hob power while cooking	1	kW	Assumed	

Parameter	Value	Unit	Quality	Src
Laundry loads per person	1	per week	Reported	[29]
Washer energy per cycle (cold wash)	0.3	kWh/cycle	Assumed	
Tumble-dryer heater + drum power	3	kW	Assumed	
Drying time per load	12	min	Reported	[29]
Water per wash	60	L/cycle	Assumed	
Drinking water	2	L/person/day	Reported	[29]
Basin, cooking & dish-washing water	15	L/person/day	Assumed	
Toilet flushing (seawater)	30	L/person/day	Assumed	[13]
Paper used per semester (50 printed + 50 written)	100	sheets	Reported	[29]
Semesters per year	2	no.	Assumed	
Mass of one A4 80 g/m2 sheet	5	g	Derived	
Share of used paper recycled	0.5	fraction	Assumed	[29]
MTR commute distance (18 km x 2 x 6 days)	216	km/week	Reported	[29]
Weeks of commuting per year	48	weeks	Assumed	[29]
Other MTR trips (22 km x 2 x 3 per month)	132	km/month	Reported	[29]
Months per year in Hong Kong	11	months	Reported	[29]
MTR rail emission intensity	0.0723	kg CO_2e/p-km	Published	[17]
Great-circle distance uplift	1.08	-	Published	[18]
Long-haul economy, with RF	0.11704	kg CO_2e/p-km	Published	[18]
Short-haul economy, with RF	0.1258	kg CO_2e/p-km	Published	[18]
Mixed (medium meat-eater) diet	5.63	kg CO_2e/day	Published	[19]
Real discount rate for NPV and cost of saved energy	0.04	fraction/yr	Assumed	
Annual decline in savings of hardware EMOs (EAC 7.5.4(b))	0.01	fraction/yr	Assumed	[1]
Share of tumble-drying replaced by air-drying	0.8	fraction	Assumed	
Kitchen lights on with occupancy sensors	8	h/day	Assumed	
Sha Tin population (land-based non-institutional), 2025	687300	persons	Published	[33]
Sha Tin domestic households, 2025	248200	households	Published	[33]
Sha Tin average domestic household size, 2025	2.7	persons/household	Published	[33]
Average living space per person in public rental housing	14.1	m2/person	Published	[34]
Average household size in public rental housing	2.6	persons/household	Published	[34]
Housing Authority public rental households in Sha Tin	77100	households	Published	[35]
Sha Tin subsidised-sale flats (occupied), 2021 Census	61819	units	Published	[37]
Sha Tin private permanent housing (occupied), 2021 Census	104280	units	Published	[37]
Median floor area per household - public housing	35	m2	Published	[36]
Median floor area per household - private housing	47	m2	Published	[36]
Hong Kong residential electricity, 2021	13184	GWh/yr	Published	[38]
Hong Kong population, 2021 Census	7413070	persons	Published	[38]
Hong Kong domestic households, 2021 Census	2674161	households	Published	[38]
CLP Power residential sales, 2025	9966	GWh/yr	Published	[39]
CLP Power residential accounts, 2025	2528522	accounts	Published	[39]

Parameter	Value	Unit	Quality	Src
Share of district residents heating water and cooking with gas (illustrative)	0.5	fraction	Assumed	
Typical HK household electricity	275	kWh/month	Published	[7]
Persons in typical household	3	persons	Published	[7]
Median per-capita floor area (2021 Census)	16	m2/person	Published	[23]
HK domestic fresh water use	120.9	L/person/day	Published	[13]
HK per-capita GHG emissions (2023)	4.58	t CO_2e/person	Published	[10]
BEC 2021 max lighting power density - dormitory	6.2	W/m2	Published	[24]

Appendix D - EE-EAes (V.2) form entries

The submitted file 2_EE-EAes2018 (V.0)_SEE6115-8115_COMPLETED.xls is the official EMSD form with the entries below. The form is a macro-enabled workbook: its sheets are hidden until it is opened in Excel with macros enabled, and its macros re-hide them when it is saved. The entries were therefore written directly into the form's cells and tick-boxes, so the macros, tick-box behaviour, protection and layout remain exactly as issued. The form's own summary cells were updated to match:

- total energy (Part 2 item 3) and EUI (item 4);
- the Part 3 category and type counts.

CBSI and lift & escalator items are marked N/A or left blank, as explained in the Executive Summary.

Form sheet	Cell / control	Field	Value entered
Part 1(A)	C15	Name of building	Room 1003B, 10/F, Hall 13, CityU Lee Shau Kee Student Residence Village (household energy audit)
Part 1(A)	C16	Address of building	Lee Shau Kee Student Residence Village, Ma On Shan, Sha Tin District, New Territories, Hong Kong
Part 1(A)	H17	1) Date of commencement of energy audit	01/01/2026
Part 1(A)	H19	2) Date of completion of energy audit	11/10/2026
Part 1(A)	G21	3) Energy Audit Form issued on	N/A
Part 1(A)	J21	3) Energy Audit Form expired on	N/A
Part 1(A)	H23	4) Energy Audit Report reference no.	SEE8115-HEA-2026-01
Part 1(A)	tick-box 124	5) Import / export energy from/to other building?	Tick 'No'
Part 1(A)	E27	5) Building(s) importing / exporting energy	N/A
Part 1(A)	H27	5) Building(s) importing / exporting energy	N/A
Part 1(A)	J27	5) Building(s) importing / exporting energy	N/A
Part 1(B)(I)	H7	1)(b) Portion of the building entity being common area (%)	0
Part 1(B)(I)	H8	1)(c) No. of blocks of the building entity	1
Part 1(B)(I)	H9	2) Total internal floor area (m2)	15
Part 1(B)(I)	H10	3) No. of floors of the building entity	1
Part 1(B)(I)	tick-box 35	4) Major type of building facade	Tick 'Non-curtain wall'
Part 1(B)(I)	H12	5) Date of issue of occupation approval	Q3 2024 (residence opened; occupation-approval date not published)
Part 1(B)(I)	tick-box 12	6) Type of central air-conditioning provided	Tick 'Not applicable'
Part 1(B)(I)	H18	7) Average weekly operating hours - row 18	N/A
Part 1(B)(I)	I18	7) Daily average no. of occupants - row 18	N/A
Part 1(B)(I)	H19	7) Average weekly operating hours - row 19	N/A
Part 1(B)(I)	I19	7) Daily average no. of occupants - row 19	N/A
Part 1(B)(I)	H20	7) Average weekly operating hours - row 20	N/A
Part 1(B)(I)	I20	7) Daily average no. of occupants - row 20	N/A
Part 1(B)(I)	H21	7) Average weekly operating hours - row 21	N/A
Part 1(B)(I)	I21	7) Daily average no. of occupants - row 21	N/A
Part 1(B)(I)	H22	7) Average weekly operating hours - row 22	N/A
Part 1(B)(I)	H23	7) Average weekly operating hours - row 23	N/A

Form sheet	Cell / control	Field	Value entered
Part 1(B)(I)	I23	7) Daily average no. of occupants - row 23	N/A
Part 1(B)(I)	D60	8)(f) Others (please specify)	N/A - no CBSI-served area in the audited unit
Part 2	E2	1) Annual electricity, past 1st 12-month (kWh/annum)	1219.6
Part 2	G2	1) Past 2nd 12-month	N/A
Part 2	I2	1) Past 3rd 12-month	N/A
Part 2	E5	2) Other energy, past 1st 12-month (MJ/annum)	0
Part 2	G5	2) Past 2nd 12-month	N/A
Part 2	I5	2) Past 3rd 12-month	N/A
Part 2	C14	5) Monthly EUI - month 1 (Oct)	32.23
Part 2	E14	5) Monthly EUI - month 2 (Nov)	16.6
Part 2	G14	5) Monthly EUI - month 3 (Dec)	16.87
Part 2	I14	5) Monthly EUI - month 4 (Jan)	16.87
Part 2	C16	5) Monthly EUI - month 5 (Feb)	16.06
Part 2	E16	5) Monthly EUI - month 6 (Mar)	16.87
Part 2	G16	5) Monthly EUI - month 7 (Apr)	16.6
Part 2	I16	5) Monthly EUI - month 8 (May)	32.23
Part 2	C18	5) Monthly EUI - month 9 (Jun)	31.96
Part 2	E18	5) Monthly EUI - month 10 (Jul)	32.23
Part 2	G18	5) Monthly EUI - month 11 (Aug)	32.23
Part 2	I18	5) Monthly EUI - month 12 (Sep)	31.96
Part 2	C20	6) Breakdown - air-conditioning (MJ/annum)	3234
Part 2	E20	6) Breakdown - lighting (MJ/annum)	420
Part 2	G20	6) Breakdown - lift & escalator (MJ/annum)	N/A
Part 2	I20	6) Breakdown - others (MJ/annum)	736
Part 2	I22	7) Energy supply from CBSI to building's units (%)	N/A
Part 2	I24	8) Energy bill reference month	30/09/2026
Part 3	A9	EMO slot A - reference	EMO-01
Part 3	tick-box linked to P9	EMO slot A - category	Tick Category I
Part 3	tick-box linked to U9	EMO slot A - type	Tick AC
Part 3	H9	EMO slot A - description	Raise the AC set-point from 25 °C to 26 °C and keep the 20 W desk fan running for air movement. Saves 24 kWh/yr, HK$34/yr, 8 kg CO_2e/yr; cost HK$0; payback immediate.
Part 3	A11	EMO slot B - reference	EMO-02
Part 3	tick-box linked to P11	EMO slot B - category	Tick Category I
Part 3	tick-box linked to U11	EMO slot B - type	Tick AC

Form sheet	Cell / control	Field	Value entered
Part 3	H11	EMO slot B - description	Use the AC's built-in timer to switch off 2 h before waking in hot months (12 to 10 h/day). Saves 100 kWh/yr, HK$140/yr, 34 kg CO_2e/yr; cost HK$0; payback immediate.
Part 3	A13	EMO slot C - reference	EMO-04
Part 3	tick-box linked to P13	EMO slot C - category	Tick Category I
Part 3	tick-box linked to U13	EMO slot C - type	Tick AC
Part 3	H13	EMO slot C - description	Cool-season 'fan first': no AC cooling in December-February (HKO mean 16.5-18.2 °C); use the desk fan and open window, Dry mode only on humid days. Saves 107 kWh/yr, HK$150/yr, 36 kg CO_2e/yr; cost HK$0; payback immediate.
Part 3	A15	EMO slot D - reference	EMO-05
Part 3	tick-box linked to P15	EMO slot D - category	Tick Category I
Part 3	tick-box linked to S15	EMO slot D - type	Tick LG
Part 3	H15	EMO slot D - description	Use the desk task lights instead of both ceiling battens for 3 of the 6 lit hours. Saves 44 kWh/yr, HK$62/yr, 15 kg CO_2e/yr; cost HK$0; payback immediate.
Part 3	A17	EMO slot E - reference	EMO-06
Part 3	tick-box linked to P17	EMO slot E - category	Tick Category I
Part 3	tick-box linked to W17	EMO slot E - type	Tick Others
Part 3	H17	EMO slot E - description	Shorten showers from 5 to 4 minutes (phone timer or a 4-minute song). Saves 127 kWh/yr, HK$178/yr, 43 kg CO_2e/yr; cost HK$0; payback immediate.
Part 3	A19	EMO slot F - reference	EMO-08
Part 3	tick-box linked to P19	EMO slot F - category	Tick Category I
Part 3	tick-box linked to T19	EMO slot F - type	Tick EL
Part 3	H19	EMO slot F - description	Laptop power management: lower brightness, sleep after 10 min idle, battery-saver mode. Saves 28 kWh/yr, HK$40/yr, 10 kg CO_2e/yr; cost HK$0; payback immediate.
Part 3	A21	EMO slot G - reference	EMO-10
Part 3	tick-box linked to P21	EMO slot G - category	Tick Category I
Part 3	tick-box linked to W21	EMO slot G - type	Tick Others
Part 3	H21	EMO slot G - description	Air-dry laundry on a rack by the open window instead of tumble-drying (keep the dryer for bulky items in humid spells). Saves 50 kWh/yr, HK$70/yr, 17 kg CO_2e/yr; cost HK$0; payback immediate.
Part 3	A23	EMO slot H - reference	EMO-11
Part 3	tick-box linked to Q23	EMO slot H - category	Tick Category II

Form sheet	Cell / control	Field	Value entered
Part 3	tick-box linked to U23	EMO slot H - type	Tick AC
Part 3	H23	EMO slot H - description	Washroom exhaust fan: replace 24 h running with an occupancy / humidistat run-on control (~8 h/day). Saves 175 kWh/yr (household share 88), HK$246/yr, 60 kg CO_2e/yr; cost HK$800; payback 3.2 yr; NPV HK$1,117.
Part 3	A25	EMO slot I - reference	EMO-12
Part 3	tick-box linked to Q25	EMO slot I - category	Tick Category II
Part 3	tick-box linked to U25	EMO slot I - type	Tick AC
Part 3	H25	EMO slot I - description	Kitchen AC: 7-day timer + occupancy interlock, set-point locked at 25.5 °C (8 to 5 h/day hot, 4 to 2 h/day cool). Saves 524 kWh/yr (household share 33), HK$737/yr, 178 kg CO_2e/yr; cost HK$1,000; payback 1.4 yr; NPV HK$4,733.
Part 3	A27	EMO slot J - reference	EMO-14
Part 3	tick-box linked to Q27	EMO slot J - category	Tick Category II
Part 3	tick-box linked to W27	EMO slot J - type	Tick Others
Part 3	H27	EMO slot J - description	Fit a WELS Grade-1 low-flow showerhead (9 to 7 L/min) in the shared washroom. Saves 282 kWh/yr (household share 141), HK$397/yr, 96 kg CO_2e/yr; cost HK$250; payback 0.6 yr; NPV HK$1,482.
Part 3	A29	EMO slot K - reference	EMO-18
Part 3	tick-box linked to R29	EMO slot K - category	Tick Category III
Part 3	tick-box linked to W29	EMO slot K - type	Tick Others
Part 3	H29	EMO slot K - description	Replace 24.4 kW instantaneous shower heaters with central air-source heat-pump water heating (COP about 3.5) and storage. Saves 910 kWh/yr (household share 455), HK$1,280/yr, 310 kg CO_2e/yr; cost HK$20,000; payback 15.6 yr; NPV HK$-6,626.
Part 3	A31	EMO slot L - reference	EMO-21
Part 3	tick-box linked to R31	EMO slot L - category	Tick Category III
Part 3	tick-box linked to W31	EMO slot L - type	Tick Others
Part 3	H31	EMO slot L - description	Expand rooftop solar PV under the CLP Feed-in Tariff (the village already has 110 panels). Viability statement only (EAC 7.4.4(d)); no saving claimed. Other EMOs (03, 07, 09, 13, 15, 16, 17, 19, 20) are in the report's EMO register (21 EMOs).

Appendix E - Completed NELC Home Energy Audit Tool

A copy of the course tool (1_SEE6115-8115_Home Energy Audit Tool_COMPLETED.xlsx) was completed. The values were written directly into the tool's input cells, tick-boxes and answer boxes, so its navigation buttons, logo, drop-down lists and formulas are unchanged and the original file is untouched. The tool uses US units, so the following conversions were applied:

- 15 m² = 161 ft²;
- 25 °C = 77 °F;
- 24.4 kW = 83,256 Btu/h;
- window dimensions in inches.

Sheet	What was filled
Main Page	Project details, occupant contact (no phone/e-mail), electricity price HK$1.406/kWh, home details (2024, 161 ft², 1 bedroom, shared bathroom, unit on 10/F)
Survey (NELC homeowner survey)	Home details (time lived there, residents by age, year built, renovations, favourite room, concerns), values and engagement scales, energy quiz, indoor-air, heating/cooling, moisture and fire-safety checks, habits, planned improvements, monitoring and renewables, and the 1-10 satisfaction ratings: 68 tick-boxes and 15 answer boxes
NELC Questionnaire V1	Indoor-air-quality questions (air quality, odours, asbestos, VOC and lead paint, humidifiers, symptoms, detectors, radon): 9 tick-boxes
Questionnaire	Occupants by age, thermostat (standard), cooling set-point 77 °F, appliance use, laundry (1 cold/cold load, 1 dryer load), weather-stripping, water heating, AC 8 h/day average × 12 months, notes
Bedrooms	Renamed 'Room 1003B'; 2 LED fixtures; single-pane aluminium windows with drapes and dimensions; laptops; 1 portable fan; absent equipment entered as 0; notes
Bathrooms	Renamed 'Shared washroom (4 users)'; LED downlights; absent equipment entered as 0; notes on heater and 24 h fan
Kitchen	LED lighting (6 × 15 W, 16 h/day); electric stove, no oven, water chiller/heater, 2 refrigerators (2023), microwave; absent small appliances entered as 0; notes
Misc Areas	Tankless electric water heater (indoors, 0.99, 83,256 Btu/h), room AC 11,400 Btu/h EER 9.6 (2024), no heating, laundry (electric), notes
Home Exterior	Orientation NW, 1 storey, rectangle, concrete construction, notes on MiC building
Appliance Dataplate Info	10 appliances with manufacturer, model, year and key ratings
Other Appliances	Laptops and phone chargers with use and wattage

Answers on attitudes and satisfaction in the Survey sheet reflect the auditor's questionnaire and behaviour and should be reviewed by the auditor before submission.

Appendix F - Deliverables and reproducibility

File	Purpose
deliverables/Home_Energy_Carbon_Audit_Report.docx (and .pdf)	This report
deliverables/1_SEE6115-8115_Home Energy Audit Tool_COMPLETED.xlsx	Assignment part (1): completed NELC Home Energy Audit Tool
deliverables/2_EE-EAes2018 (V.0)_SEE6115-8115_COMPLETED.xls	Assignment part (2a): completed EMSD EE-EAes executive-summary form (open in Excel with macros enabled)
deliverables/Home_Energy_Carbon_Audit_Workbook.xlsx	Calculation workbook - 16 sheets, 1,452 live formulas, no formula errors
deliverables/figures/	The 15 charts used in this report
photos/	26 evidence photographs
audit-inputs/QUESTIONNAIRE.md	Completed questionnaire
scripts/	Python/Node scripts that rebuild the workbook, verify it against an independent model, fill both course forms and generate this report

Rebuild sequence: build_workbook.py, then recalculation, then verify_workbook.py, then fill_home_audit_tool.py and fill_eeaes_form.py, then make_report_assets.py, then build_report.js, then finalize_report.py (contents page numbers and PDF copy). Every number in this report is read from the recalculated workbook.